

Predictive Urban Water Stress Mapping and Decentralized Infrastructure Siting in Bengaluru: A GeoAI Approach



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List of Abbreviations

- BBMP – Bruhat Bengaluru Mahanagara Palike
- SPI – Standardized Precipitation Index
- LST – Land Surface Temperature
- CA (Markov) – Cellular Automata (Markov)
- ARIMA – Autoregressive Integrated Moving Average
- OLS – Ordinary Least Squares
- BWSSB – Bengaluru Water Supply and Sewerage Board
- CWSS – Cauvery Water Supply Scheme
- CGWB – Central Ground Water Board
- VIC – Variable Infiltration Capacity
- MIKE SHE – MIKE Système Hydrologique Européenne
- CHIRPS – Climate Hazards Group InfraRed Precipitation with Stations
- SRTM – Shuttle Radar Topography Mission
- SII – Soil Infiltration Index
- AHP – Analytical Hierarchy Process
- MCDA – Multi-Criteria Decision Analysis
- WCSS – Within-Cluster Sum of Squares
- MSE – Mean Squared Error
- RMSE – Root Mean Squared Error
- XGBoost – eXtreme Gradient Boost

Abstract

Water Stress in Bengaluru has been a long-term issue that has significantly intensified in contemporary times, constituted by rapid urbanisation, climatic variability and unsustainable groundwater exploitation practices. Conventional methods to combat with this issue often fails to capture the complex spatial interactions among environmental, hydrological and socio-economic factors which influence water availability. This study, therefore, features a GeoAI-enabled framework to predict the spatial distribution of Bengaluru's water stress over a ten-year period as well as suggests decentralised infrastructural intervention approaches to mitigate the water scarcity issue within the Bruhat Bengaluru Mahanagara Palike (BBMP) region.

This project integrates several geospatial datasets which represent key determinants of urban water availability, including Standardized Precipitation Index (SPI), Land Surface Temperature (LST), Land Use and Land Cover (LULC), Groundwater Depth, Borewell Density, Population and Built-Up Density, Slope Factor, as well as proximity to existing water infrastructure such as surface water bodies, pumping stations and water pipelines.

Predictive layers highlighting the spatial concentration of water stress within the BBMP region has been developed for the years of 2025 and 2035. Future predictor variables for 2035 were generated using a combination of machine learning and statistical forecasting techniques. Historical LULC datasets of BBMP for the years of 2000, 2005, 2015, 2020 and 2025 have been produced and validated using a Supervised Random Forest Classification technique and Cellular Automata (CA) Markov Model has been used to simulate the 2035 LULC scenario. Rainfall and subsequently SPI were forecasted using the Facebook Prophet time-series model and LST has been projected using the statistical ARIMA model. Groundwater Depth for the year 2035 has been estimated using Ordinary Least Squares (OLS) linear regression. Remaining dataset has been kept consistent for both the 2025 and 2035 prediction. These datasets were then utilised to train Random Forest and XGBoost machine learning algorithms for predictive urban water stress mapping and comparative assessment of model performance and spatial distribution of water stress across Bengaluru.

Finally, an unsupervised K-Means Clustering algorithm has been utilised to site decentralised water infrastructure locations across BBMP to support sustainable water resource planning alongside governmental interventions. The optimal number of infrastructure clusters was determined using the Elbow Method, Silhouette Score as well as Davies-Bouldin Index.

Preface

Bengaluru, the “Silicon Valley of India” has experienced unprecedented urban expansion, combined with groundwater depletion and decline in surface waterbodies which has intensified the city’s already stressed existing water infrastructure.

Often termed as the “City of Lakes”, Bengaluru once had an interconnected network of urban lakes and tanks which supported the city’s water demands, as well as naturally recharged groundwater. However, large-scale urbanisation meant degradation and encroachment of these waterbodies. Subsequently, Bengaluru was over-dependent on its groundwater availability to meet the water need of its population. This excessive groundwater extraction and dependence on the nearby Cauvery River have resulted in an imbalance between Bengaluru’s shrinking water availability and growing water needs. Today, several locations across Bruhat Bengaluru Mahanagara Palike (BBMP) region including Whitefield, Mahadevapura, Electronics City etc. are under severe water stress situation, with groundwater depths several meters below the average recommended level.

Urban Water Stress is a complex phenomenon resulting from the interaction among several environmental, hydrological and socio-economic variables making it difficult to map spatially. Remote Sensing, GIS and Machine Learning based approaches therefore presents an opportunity to combine such diverse datasets into a predictive framework that supports evidence-based urban planning and recommend decentralised infrastructure interventions in order to improve long-term urban water resilience.

Against this backdrop, this report aims to identify the major drivers of water stress in Bengaluru, predict water stress hotspots and provide a scientific basis for understanding future urban water stress and supporting sustainable infrastructure planning within the BBMP region. The findings are intended to contribute towards informed decision-making to strengthen the sustainability of Bengaluru’s water resources.

CHAPTER 1:

Introduction

1.1 Introduction

Freshwater stress in megacities of the Global South is critical with spatially uneven and ever-increasing urbanisation as a major consequential driver. Bengaluru, the capital of Indian state of Karnataka is a prime example of an urban centre battling with water scarcity.

Bengaluru, which once had over 1450 lakes within its administrative boundary witnessed a population growth of 8.5 million in 2011 from 1.65 million people in 1971 (Census of India). Built-up area expanded from roughly 20% to over 80% of its corresponding municipal boundaries. Between 2001 and 2011, certain wards of Bengaluru recorded over 150% to 300% population growth. Within the same time period, vegetation cover fell from around 30% to less than 2%. Bengaluru's population density rose from 2985 persons per square kilometre in 2001 to 4378 persons per square kilometre in 2011.

The Bengaluru Water Supply and Sewerage Board (BWSSB) provides water supply to BBMP region of Bengaluru, and it heavily relies on the installed Cauvery Water Supply Scheme (CWSS) capacity of 1440 million litres per day (MLD). This supply however, is insufficient because Bengaluru's population requires approximately 2600 MLD of freshwater.

Groundwater has therefore historically served as the default relief valve for this water deficit but years of unregulated extraction has caused a contemporary aquifer exhaustion. 1392 MLD is extracted annually from groundwater but only about 148 MLD is naturally replenished. This massive distinction has resulted in Bengaluru's groundwater levels reaching a critical stage. Certain regions of Bengaluru have groundwater depth over 60 metres below ground level (mbgl). Out of 14781 borewells under BWSSB, 6997 have ceased to yield water as of early 2024.

The water stress is further accelerated with the progressive loss of surface waterbodies which amplified the feedback between urban expansion and groundwater depletion. According to an IISc. Report, in half a century since 1973, Bengaluru lost nearly 88% of its vegetation cover and 79% of its waterbodies. The thermal consequence of this landscape transformation compounded water stress through a Urban Heat Island (UHI) effect. This can be validated through a research article which stated that Bengaluru's Land Surface Temperature (LST) rose by 15.13°C over the last thirty years. The geographies of peak built-up density and high LST spatially coincide with zones of acute groundwater depletion. These zones include Whitefield, Electronic City, Sarjapur Road, Hebbal etc.

Knowing these factors, a spatially heterogeneous Urban Water Stress Index is developed in this study, for the year 2025 and a predicted 2035 index, with the help of ensemble machine learning classifiers Random Forest and XGBoost. The machine learning algorithms have been fed with 11 primary drivers of water stress observed in Bengaluru. The index is visualised into three ordinal classes: ‘Low Stress’, ‘Moderate Stress’ and ‘High Stress’. The outputs are intended to create a ward-prioritised decentralised intervention planning, identifying zones where distributed infrastructure can yield the highest water security returns for the most vulnerable urban populations.

1.2 Problem Statement

Bengaluru suffers from a chronic and deepening urban water stress, with groundwater extraction reaching approximately 1392 MLD against a natural recharge of only 148 MLD and a total freshwater deficit of nearly 500 MLD persisting daily despite Cauvery surface supply. All six groundwater assessment units within BBMP are over-exploited as per CGWB reports and half of the city’s existing borewells have run dry.

Despite the scale of this water crisis, limited study exists that provides a ward-level multi-dimensional mapping of water stress severity that integrates biophysical drivers like thermal load and precipitation with infrastructural realities such as borewell density, pumping stations and waterline infrastructure. Existing research frameworks largely focuses on groundwater and related causes whilst excluding several interacting parameters that drive the compound water stress scenario. The absence of such a composite spatial framework is addressed in this report through a GeoAI-driven ensemble machine learning framework that combines every factor into an Urban Water Stress Index. This report further proposes ideal locations for decentralised water infrastructure allocation.

1.3 Research Gap

Existing studies have predominantly focused on Bengaluru’s groundwater availability and depletion owing to the city’s heavy dependence on groundwater resources. Water stress is a multidimensional phenomenon influenced by several environmental, infrastructural, hydrological as well as socio-economic variables such as rainfall variability, surface temperature, population concentration and accessibility to existing water infrastructure. Therefore, there exists a need for a comprehensive study which incorporates these diverse determinants into a unified spatial modelling framework.

Furthermore, it has been observed that previous studies primarily relied on conventional spatial analysis or descriptive assessments based on existing governmental observations and datasets. Thus, the application of a GeoAI framework using ensemble machine learning models like Random Forest and XGBoost for predictive water stress modelling is expected to provide a more detailed and reproducible analysis for Bengaluru's water stress scenario as well as future similar studies.

Moreover, an additional limitation observed in existing literature is that majority of studies conclude with the identification and mapping of water stress hotspots. The translation of water stress assessments into an actionable decentralized water infrastructure recommendations remain insufficiently explored. This project therefore addresses this gap by employing K-Means Clustering to identify suitable locations for decentralised water infrastructure interventions.

1.4 Research Questions

- a. What are the primary environmental, hydrological, infrastructural and socio-economic drivers influencing the spatial distribution of Bengaluru's water stress?
- b. How effectively can GeoAI-based machine learning models, especially Random Forest and XGBoost predict water stress for the years 2025 and 2035?
- c. How are projected climatic variations and urban growth expected to influence the future spatial distribution and intensity of water stress in Bengaluru?
- d. How can K-Means clustering be utilized to identify optimal decentralised water infrastructure interventions across Bengaluru?
- e. Can an integrated geospatial decision-support framework that combines predictive water stress modelling with infrastructure siting contribute to a sustainable and evidence-based long-term urban water resilience in Bengaluru?

1.5 Research Objectives

- a. To generate future forecast scenarios of environmental and hydrological variables of Bengaluru's water stress using GeoAI-based stochastic and statistical modelling.
- b. To produce predictive urban water stress hotspot maps for the years 2025 and 2035 of Bengaluru using Random Forest and XGBoost models and evaluate their performance.

- c. To identify suitable locations for decentralised water infrastructure allocation through a K-Means clustering algorithm and develop a spatial decision-support framework for sustainable urban water resource planning.

CHAPTER 2:

Literature Review

2.1. Water Stress of Bengaluru: Background and Context

The condition of an acute water stress scenario occurs when the demand for water supply largely exceeds available quantity or quality of accessible water resources. Water stress has constituted itself as a critical challenge in rapidly urbanizing cities, particularly in developing countries like India, where population growth, land-use transformation and climate variability exert increasing pressure on already finite water resources.

The city of Bengaluru exemplifies this challenge, with several studies consistently reporting a widening gap between rising water demands and limited available supply, driven by rapid urbanization, declining groundwater levels, shrinking urban lakes and increasing dependency on external water sources ([Krishna Raj, 2013](#); [Ramachandra et al., 2016](#); [Moharaj et al., 2025](#)). The city's expansion has substantially altered its hydrological landscape by replacing natural recharge zones with impervious surfaces and thereby reducing groundwater replenishment potential ([Ramachandra et al., 2016](#); [Bhoomika et al., 2024](#)).

Although large-scale infrastructural development projects continue to augment water supply from the Cauvery River, existing literature indicates that nearly 49% of the city's total water supply by the Bengaluru Water Supply and Sewerage Board (BWSSB) still remains heavily dependent on groundwater extraction to sustain domestic, industrial and commercial activities ([JICA, 2017](#); [WELL Labs, 2024](#)).

2.2 Existing Water Stress Assessment Framework: A Review

Contemporary researchers have extensively employed Remote Sensing and GIS techniques to monitor environmental changes associated with water stress. Multi-temporal satellite datasets, particularly from the NASA's Landsat series, have been used to evaluate changes in land use, vegetation health and surface water dynamics through indices such as NDVI, NDWI and MNDWI ([Malunda, 2019](#); [Ramachandra et al., 2016](#)). These studies demonstrate a strong positive relationship between urban expansion, declining vegetation cover and shrinking water bodies. Hydrological investigations further reveal significant modifications in Bengaluru's water balance, with anthropogenic recharge from leaking water supply and wastewater infrastructure contributing more recharge than natural rainfall infiltration ([Tomer et al., 2020](#); [Brauns et al., 2022](#)).

Process-based models such as VIC, MIKE SHE and Groundwater-Budget models have provided valuable insights into runoff generation, recharge dynamics and water availability under varying climatic conditions ([Joshi et al., 2024](#); [Tomer et al., 2020](#)). The Variable

Infiltration Capacity (VIC) model has been particularly employed for large-scale water balance simulations, owing to its ability to simultaneously resolve surface runoff, baseflow and actual evapotranspiration under varying land cover and climatic forcing conditions ([Liang et al., 1994](#); [Joshi et al., 2024](#)). The Priestley–Taylor (P–T) algorithm is another widely adopted method used for estimating evapotranspiration (ET) and potential evapotranspiration (PET) in hydrological and water resource studies. The method estimates evapotranspiration primarily from net radiation and temperature-based energy balance relationships, reducing the dependence on extensive meteorological inputs required by more complex models such as the Penman–Monteith equation. Owing to its simplicity and reliability, the P–T algorithm has been extensively applied in drought assessment, water balance modelling, agricultural monitoring and climate change studies. ([Abderrahmane et al., 2014](#)).

Several government-based studies, conducted particularly by the Central Ground Water Board (CGWB) and Bengaluru Water Supply and Sewerage Board (BWSSB) have investigated Bengaluru’s water stress with heavy focus on groundwater dynamics and existing borewells situation. These studies have consistently concluded groundwater depletion as the major driver of Bengaluru’s water crisis as the city lacks any reliable alternative mechanisms for water supply. These studies have also emphasized the importance of having a consistent city-wide water supply infrastructure. Localities further away from the Cauvery water transmission pipelines and pumping stations exhibit greater reliance on groundwater and therefore experiences higher levels of water stress. ([Groundwater Outlook of Bengaluru City, April 2025](#))

2.3 Assessment of Hydro-Climatic Water Stress Monitoring Indices

Various drought and water stress indicators have been employed to assess water scarcity by integrating both climatic and socio-economic dimensions of water availability. Meteorological drought indicators such as the Standardized Precipitation Index (SPI) and Effective Drought Index (EDI) utilize long-term precipitation records to quantify rainfall deficits and drought severity across different temporal scales ([Singh et al., 2022](#)). However, these indices do not account for atmospheric water demand. To address this limitation, the Standardized Precipitation Evapotranspiration Index (SPEI) incorporates both precipitation and potential evapotranspiration (PET), making it particularly suitable for evaluating climate change impacts on water resources ([Borah, 2025](#)). Statistical techniques such as the Shapiro–Wilk and Jarque–Bera tests are often used to validate the underlying data distributions for SPEI calculations ([Dilling et al., 2019](#)).

Beyond climatic indicators, several studies have incorporated socio-economic measures to capture human-induced pressures on water resources. The Water Exploitation Index (WEI) evaluates the ratio of water withdrawals to available resources, while the De Martonne Aridity Index (MA) combines temperature and precipitation to assess regional aridity conditions ([Rathore et al., 2020](#)). Additionally, remotely sensed indicators such as the Vegetation Condition Index (VCI), derived from MODIS imagery, have been widely used to monitor vegetation stress associated with drought and groundwater depletion ([Drought Indicators Handbook – Integrated Drought Management Programme](#))

Studies utilizing these indicators consistently demonstrate that urban water stress is not solely driven by precipitation deficits but is significantly intensified by rapid urbanization, population growth and excessive groundwater abstraction. The integration of climatic, hydrological and socio-economic indicators provides a comprehensive understanding of water stress dynamics by capturing both supply-side scarcity and demand-side pressures. However, a major limitation of existing studies is their reliance on regional or state-level assessments using relatively coarse-resolution datasets.

2.4 Site Selection for Decentralised Infrastructural Interventions

Recent advancements in artificial intelligence, IoT-enabled monitoring systems and predictive analytics have demonstrated considerable potential for improving water resource management through demand forecasting, leakage detection and operational optimization ([Iyer & Giri, 2020](#); [Moharaj et al., 2025](#)). Similarly, GIS-Based MCDA-AHP techniques have been widely used to identify suitable locations for rainwater harvesting structures, artificial recharge systems and decentralized water infrastructure interventions ([Tholiya & Chaudhary, 2021](#)).

Furthermore, recent research is highly exploring the potential of unsupervised machine learning techniques for spatial planning and resource allocation as they are able to identify and group regions showcasing similar environmental and socio-economic characteristics. K-Means clustering, among them, is widely used and accepted due to its computational efficiency and ability to generate homogeneous spatial clusters from multi-dimensional datasets. ([Hosseini et al., 2025](#), [Abdolahi et al., 2025](#))

2.5 Research Gaps and Future Research Directions

Despite these advancements, several critical research gaps remain. Much of the existing literature focuses either on monitoring current water stress conditions through remote sensing and hydrological modelling or on evaluating individual water management interventions in

isolation. Limited research has been performed on integrating hydrological, environmental, socioeconomic and geospatial variables within a unified predictive framework capable of identifying future urban water stress hotspots. Furthermore, while machine learning and artificial intelligence have been increasingly adopted for operational water management, their application in spatially explicit urban water stress prediction remains relatively underexplored, particularly in Indian metropolitan contexts. Existing studies also tend to emphasize water stress assessment without simultaneously addressing where decentralized infrastructure interventions should be strategically implemented to mitigate future risks.

Therefore, a significant opportunity exists to develop a GeoAI-driven framework that not only predicts the spatial distribution of urban water stress but also identifies optimal locations for decentralized water infrastructure interventions. By integrating remote sensing datasets, hydrological indicators, machine learning techniques and GIS-based suitability analysis, the proposed study seeks to bridge the gap between water stress assessment and actionable planning, which is particularly relevant due to the contemporary water bankruptcy experienced by several Indian metro cities, including Bengaluru ([UNU-INWEH Report: Madani, K., 2026](#)).

CHAPTER 3:

Study Area

3.0 Study Area Selection and its Justification

The study area (Figure 1) comprises the administrative boundary of the Bruhat Bengaluru Mahanagara Palike (BBMP), which forms the urban core of Bengaluru, Karnataka, India. Located on the southeastern Deccan Plateau, the city extends approximately between latitudes 12°50'N–13°08'N and longitudes 77°27'E–77°47'E, covering an area of nearly 709 km². Bengaluru is situated at an average elevation of approximately 920 m above mean sea level, with terrain elevations ranging from about 840 m to 960 m. The city is characterized by an undulating plateau landscape traversed by a prominent north-northeast to south-southwest ridge system that strongly influences local drainage patterns and watershed boundaries.

Geologically, Bengaluru is underlain by Precambrian granitic gneisses, migmatites and doleritic intrusions, overlain by weathered regolith and predominantly red loamy soils with moderate infiltration capacity. The city experiences a tropical savannah climate (Köppen Aw) with moderate temperatures throughout the year owing to its elevated location. Mean annual temperatures are approximately 24°C, while annual rainfall generally ranges between 850 mm and 1,000 mm. Rainfall is primarily received during the Southwest Monsoon (June–September), supplemented by the Northeast Monsoon (October–December). Increasing climatic variability, irregular rainfall distribution and recurrent drought episodes have intensified urban water stress in recent years.

Hydrologically, Bengaluru lacks a major perennial river within its municipal boundary and historically depended on an interconnected system of man-made lakes, tanks and drainage channels for water storage, flood moderation and groundwater recharge. The city is organized around three principal valley systems: the Vrishabhavathi Valley, the Koramangala–Challaghatta Valley and the Hebbal Valley. These valleys collectively form the city's primary surface drainage network and ultimately drain toward the Cauvery and South Pennar river basins. Groundwater occurs mainly within weathered and fractured hard-rock aquifers characterized by low storage capacity and limited recharge potential, making the city highly vulnerable to groundwater depletion under intensive extraction.

Rapid urbanization over the past few decades has substantially altered Bengaluru's natural hydrological regime. Expansion of impervious built-up surfaces, encroachment of drainage corridors, reduction in vegetation cover and degradation of urban lakes have reduced natural infiltration and groundwater recharge while increasing surface runoff and flood susceptibility. Simultaneously, growing population pressure and rising water demand have increased

dependence on imported Cauvery water and groundwater abstraction, contributing to recurring water scarcity across several parts of the city.

These characteristics make the BBMP region an ideal study area for applying Geographic Information Systems (GIS), Remote Sensing and GeoAI techniques to assess urban water stress. The complex interaction among land-use change, hydrological processes, climatic variability and urban infrastructure provides a suitable framework for developing predictive water stress model.

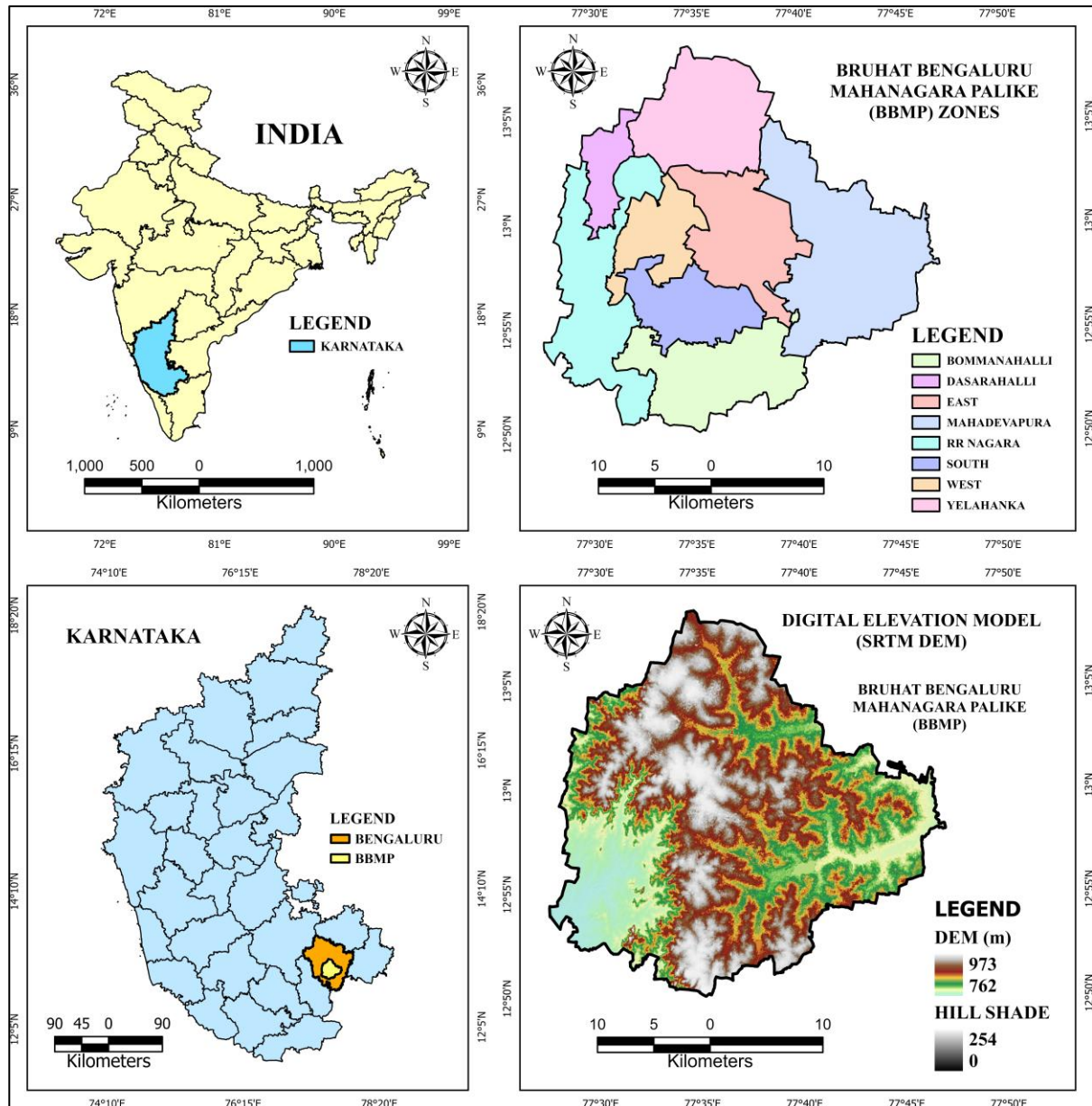


Fig. 1: Location of Study Area: Bengaluru City (BBMP) & Digital Elevation Model (m) of BBMP.

CHAPTER 4:

Data Source

4.0 Data Source

4.0.1 Rainfall and Standardized Precipitation Index (SPI)

Annual precipitation data as well as total monsoonal rainfall (June-September) of Bengaluru city (BBMP) over a 25 years period (2000-2025) has been sourced from Climate Hazards Group InfraRed Precipitation with Station (**CHIRPS**), having a native spatial resolution of 5 km. Furthermore, a spatial raster of distribution of annual rainfall over BBMP zone for the year 2025 has also been generated from CHIRPS for further processing, forecasting and computation of Standardized Precipitation Index (SPI)

4.0.2 Land Surface Temperature (LST)

Pre-monsoon Land Surface Temperature data (March-May) has been extracted from **MODIS MOD11A1 V6.1** product for the years 2000-2025 at 1 km spatial resolution. Moreover, spatial LST raster for the year 2025 has also been prepared for further data processing and analysis

4.0.3 Land Use Land Cover Classification (LULC)

Land Use Land Cover (LULC) of Bengaluru has been classified using supervised Random Forest classification using Google Earth Engine (GEE) by using NASA's **Landsat 5** satellite data for the years 2000 and 2005. For the years 2015, 2020 and 2025, **Landsat 8 OLI** satellite data have been used. Owing to BBMP'S urban landscape, three classes have been considered, namely, Built-Up, Waterbody and Vegetation. The derived LULC maps have spatial resolutions of 30 m.

4.0.4 Built-Up Density

Built-Up Density maps for years 2000, 2005, 2015, 2020 and 2025 have been prepared using a **focal statistics-based** approach over LULC maps of each year. Methodology of this process shall be explained later.

4.0.5 Elevation and Slope

Shuttle Radar Topography Mission (**SRTM**) Digital Elevation Model (DEM) of Bengaluru's BBMP area has been sourced with the help of Google Earth Engine. SRTM DEM has a spatial resolution of 30 m. Slope (in degrees) has been computed from DEM in ArcGIS Pro.

4.0.6 Soil Type Information

Soil type information for BBMP area has been extracted from ISRIC World Information's **SoilGrids v2.0** product, with a native spatial resolution of 250 m, with the help of Google Earth

Engine. Data involving sand content and clay content (both in percentages) have been utilised to prepare the Soil Infiltration Index map of Bengaluru.

4.0.7 Road and Surface Water

An open-source vector layer constituting the major road network for the entire country of India has been used from OpenStreetMap (OSM). This layer is then clipped to BBMP extent with the help of ArcGIS Pro tool.

Waterbodies for the year 2025 and 2035 have been extracted from **LULC maps** of each year respectively, where class 3 has been the ‘Waterbody’ class for both the LULC rasters.

Table 1: Data Sources

Data Layer	Source	Spatial Resolution	Year	Format
Rainfall	CHIRPS	5.5 km (Resampled to 30 m)	2000-2025	Raster
SPI	Rainfall-Derived	5.5 km (Resampled to 30 m)	2000-2025	Raster
LST	MODIS	1 km (Resampled to 30 m)	2000-2025	Raster
LULC & Built-Up Density	Landsat 5 & Landsat 8 OLI	30 m	2000, 2005, 2015, 2020, 2025	Raster
DEM	SRTM	30 m	2025	Raster
Slope	Derived	30 m	2025	Raster
Soil Type Data	SoilGrids	250 m (Resampled to 30 m)	2025	Raster
Road Network	OSM	Line	2025	Vector
Surface Water	Derived	Polygon	2025	Vector
Population	GBA	Polygon	2025	Vector
Groundwater Depth	BWSSB	Point	2012, 2016, 2024	Vector
Borewell Locations	BWSSB	Point	2024	Vector
Pumping Stations	BWSSB	Point	2024	Vector
Water Pipeline Infrastructure	BWSSB	Line	2024	Vector

4.0.8 Government Data

4.0.8.1 Population

Ward-wise population data of BBMP has been extracted as a vector KML layer from the **Greater Bengaluru Authority – GBA** portal.

4.0.8.2 Groundwater Level, Borewell Locations, Pumping Stations, & Infrastructure

The Bangalore Water Supply and Sewerage Board (**BWSSB**) portal provided vector KML layers for the following:

- a. Groundwater Depth (in mbgl – metres below ground level) data for 45 monitoring stations across BBMP.
- b. 22314 Borewell locations across BBMP area.
- c. 90 Water Pumping Stations locations across BBMP area.
- d. Water Pipeline Infrastructure vector layer of BBMP area.

CHAPTER 5:

Methodology

5.1 LULC Classification using Supervised Random Forest

Multi-temporal Land Use and Land Cover (LULC) rasters for the years 2000, 2005, 2015, 2020 and 2025 have been prepared using a supervised Random Forest method with the help of Google Earth Engine (GEE).

Random Forest is the ideal classification technique in this study due to its high classification accuracy, robustness against overfitting and capability to work with high-dimensional remote sensing data.

A total of 250 reference samples were collected for three LULC classes of BBMP, as observed through extensive study:

- Vegetation
- Built-Up
- Waterbody

The data was divided into a 70:30 ratio, where 70% of the samples (175 points) have been used for training the Random Forest model and 30% of the samples (75 points) were used for model testing, validation and accuracy assessment.

Overall Accuracy (OA) and Kappa Coefficient (K) have been computed for all. The Confusion Matrix has been generated using Python, for all the years, and visualised (Figure 2).

Table 2: Overall Accuracy (%) and Kappa Coefficient (All Years)

Year	Overall Accuracy (%)	Kappa Coefficient (K)
2000	85.78	0.7867
2005	87.56	0.8133
2015	88.89	0.8333
2020	89.78	0.8467
2025	88.00	0.8200

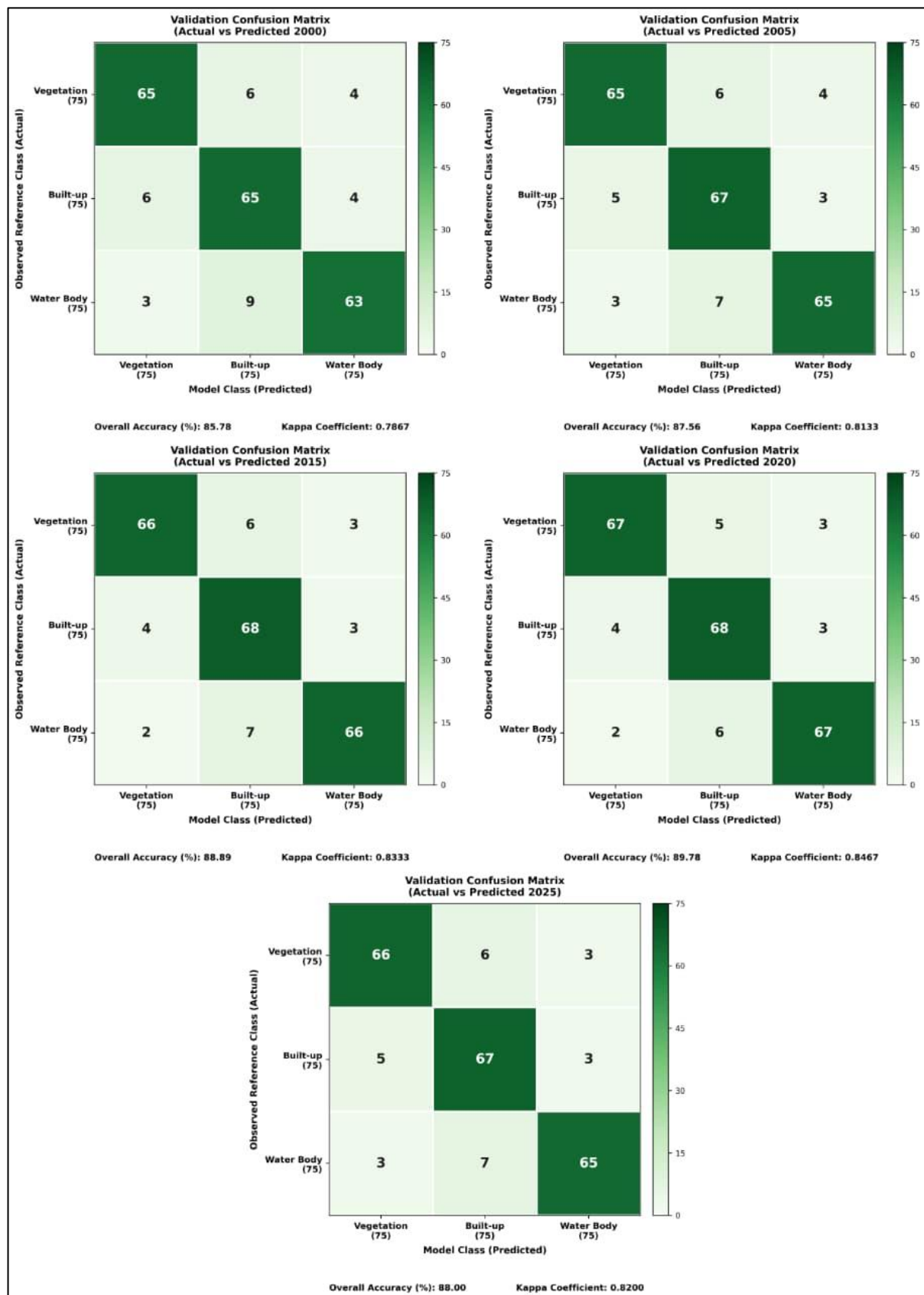


Fig. 2: Confusion Matrix (2000, 2005, 2015, 2020, 2025)

5.2 Forecasting Variables for year 2035

5.2.1 Rainfall and SPI Forecast using Facebook Prophet Model

Historical annual rainfall dataset from 2000 to 2025 (Figure 3) has been used as the input variable, on which Prophet model has been used to forecast rainfall over BBMP in 2035. Prophet Model has been selected due to its capability to model non-linear temporal trends, seasonal variability and long-term changes.

Prophet forecasting represents observed rainfall as the additive combination of trend, seasonality and residual components:

$$y(t) = g(t) + s(t) + h(t) + \varepsilon_t$$

where:

- $y(t)$ = annual rainfall at time t ,
- $g(t)$ = long-term trend component,
- $s(t)$ = seasonal component,
- $h(t)$ = effects of special events or holidays,
- ε_t = random error term.

Then, a scale factor was applied to get the spatial distribution of rainfall in 2035 over BBMP.

The scale factor was calculated as follows:

$$SF = R_{2035} / R_{2000-2025}$$

where:

- SF = rainfall scale factor,
- R_{2035} = Prophet-predicted annual rainfall for 2035,
- $R_{2000-2025}$ = mean annual rainfall during 2000–2025.

The Standardized Precipitation Index has been developed and forecasted similarly, from rainfall data to quantify meteorological wetness and drought conditions. The formula used is:

$$SPI = (P - \mu) / \sigma$$

where:

- P = annual precipitation,
- μ = long-term mean annual precipitation,
- σ = standard deviation of annual precipitation.

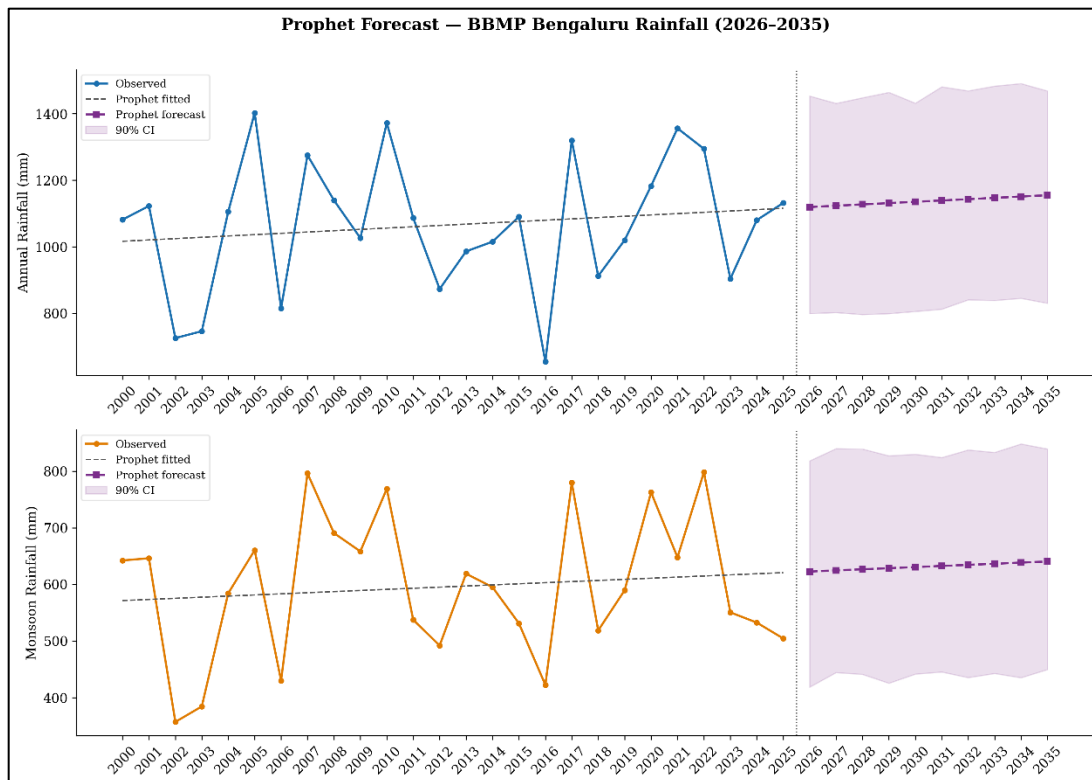
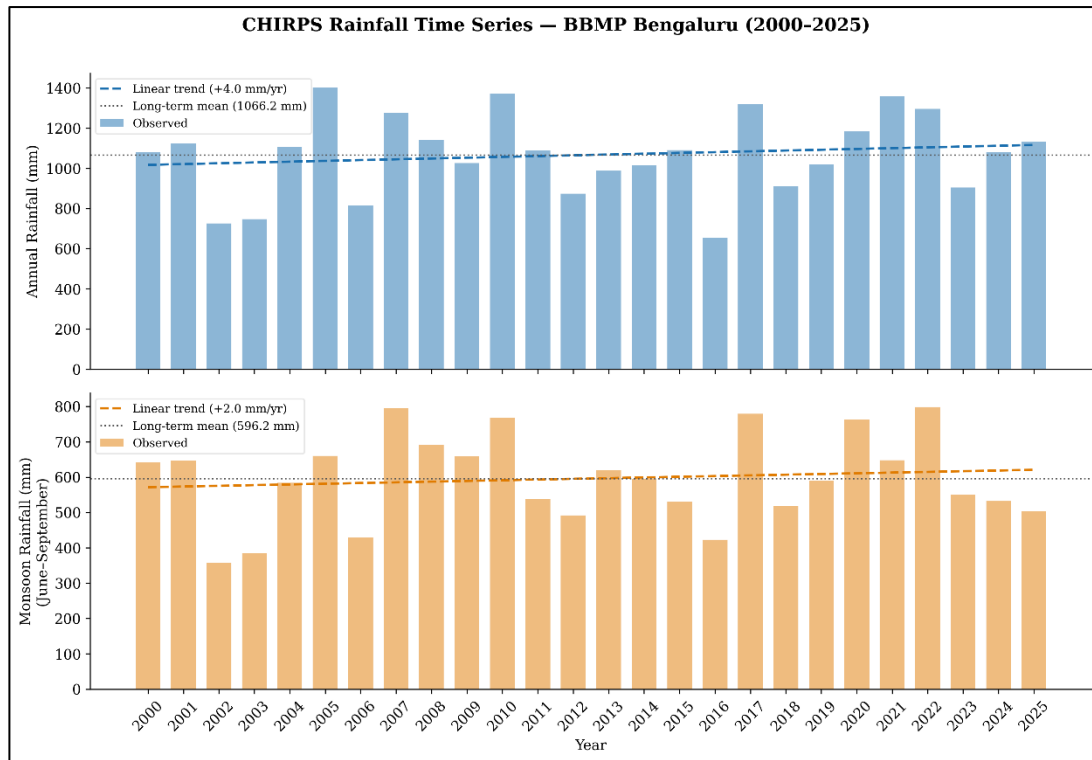


Fig. 3: (a) Rainfall in BBMP (2000-2025), (b) Prophet-forecasted rainfall (2026-2035)

5.2.2 Land Surface Temperature forecast using ARIMA Model

Annual MODIS pre-monsoon Land Surface Temperature (LST) datasets from 2020 to 2025 (Figure 4) for BBMP has been used for Autoregressive Integrated Moving Average (ARIMA) based modelling and forecasting to get spatial distribution raster of Land Surface Temperature for the year 2035. Scale Factor (SF) is applied in a similar fashion as that of rainfall variable. The ARIMA Model equation that has been employed is as follows:

$$\phi(B)(1 - B)^d Y_t = \theta(B)\varepsilon_t$$

where:

- Y_t = Land Surface Temperature at time t ,
- B = Backshift operator,
- d = Order of differencing,
- $\phi(B)$ = Autoregressive (AR) component,
- $\theta(B)$ = Moving Average (MA) component,
- ε_t = Random error term.

Table 3: Prophet and ARIMA Model: Metrics Summary

Variable	Model	MSE	RMSE	R^2
Annual Rainfall	Prophet	22500 mm ²	150 mm	0.78
LST	ARIMA	0.64° C	0.8° C	0.82

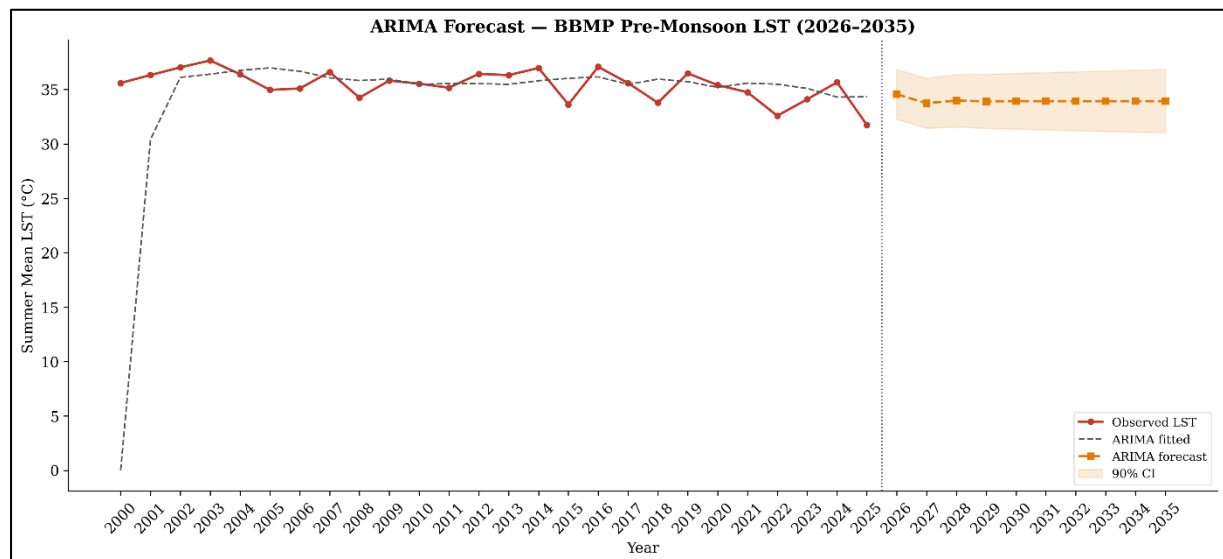


Fig. 4: Pre-Monsoon Land Surface Temperature (LST) of BBMP (2000-2025)

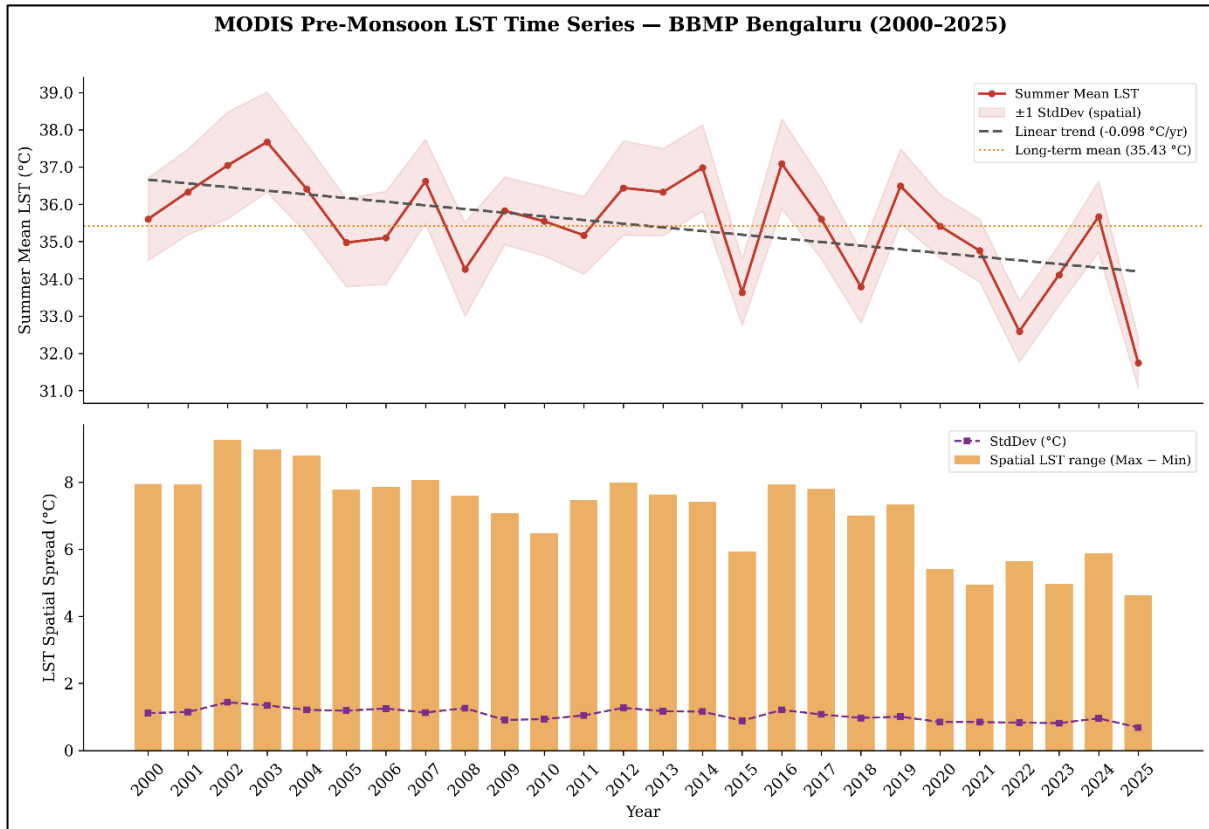


Fig. 5: ARIMA-forecasted Land Surface Temperature (LST) (2026-2035)

5.2.3 Land Use Land Cover (LULC) Forecast using CA-Markov Model

Land Use Land Cover (LULC) for the year 2035 has been simulated using the Cellular Automata-Markov approach. This model combines the temporal transition capability of the Markov Chain with the spatial neighbourhood allocation mechanism of Cellular Automata, which ensures a reliable and realistic prediction of future land use and land cover patterns.

The input parameters that have been used for the employment of this model include: Land Use & Land Cover (LULC) classified maps and Built-Up Density maps of the years 2000, 2005, 2015, 2020 and 2020, Distance to waterbody (2025), Distance to major roads and Slope (from DEM) (Figure 6). The Transition Probability between different classes is expressed as:

$$P_{ij} = n_{ij} / \sum_{j=1}^m n_{ij}$$

where:

- P_{ij} = probability of transition from land cover class i to class j ,
- n_{ij} = number of pixels changing from class i to class j ,
- m = total number of land cover classes.

The future land cover distribution is then estimated using the transition probability matrix:

$$S_{t+1} = S_t \times P$$

where:

- S_t = land cover distribution at time t ,
- P = Markov transition probability matrix,
- S_{t+1} = predicted land cover distribution

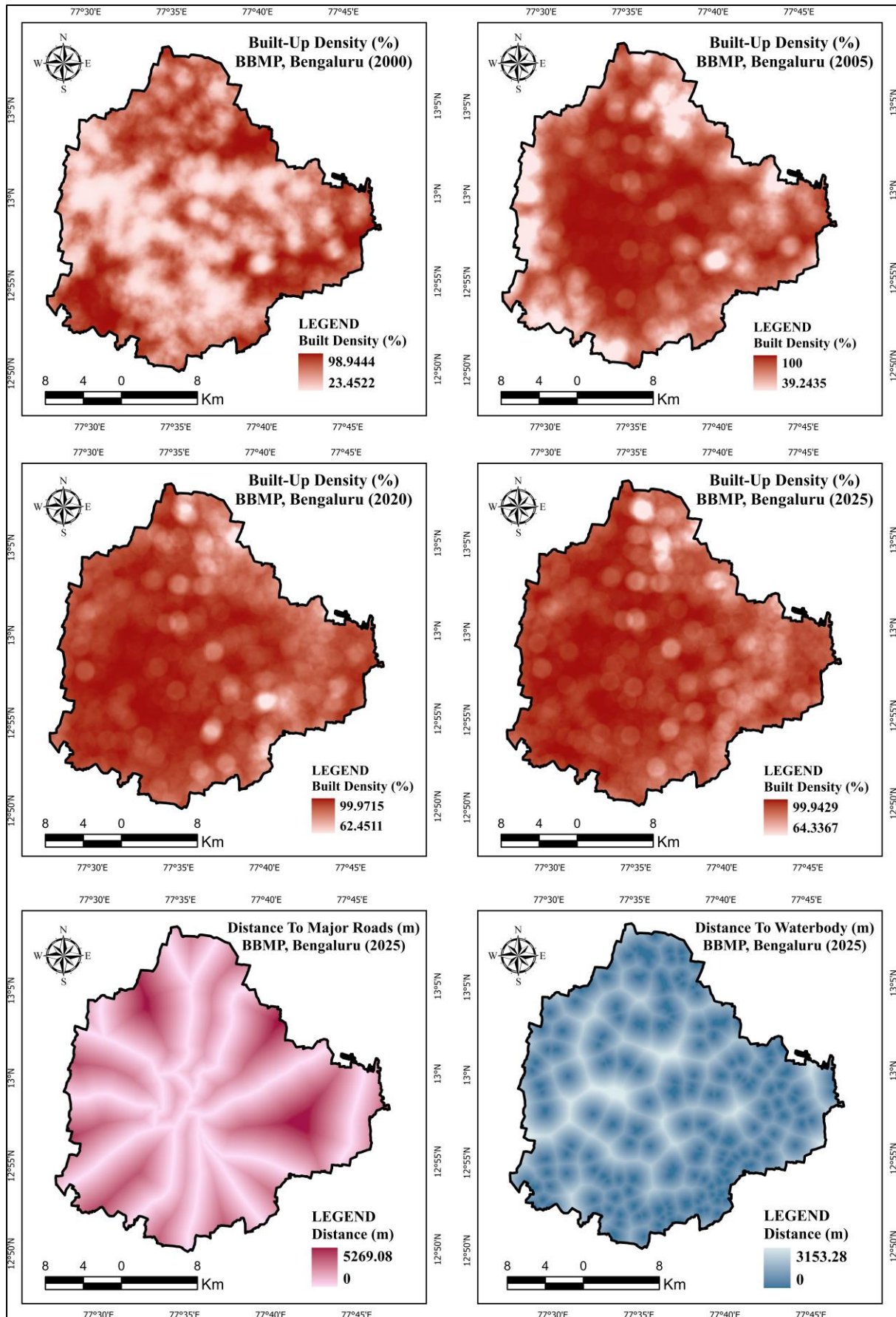


Fig. 6: Selected Input Variables for CA-Markov-Based LULC Prediction

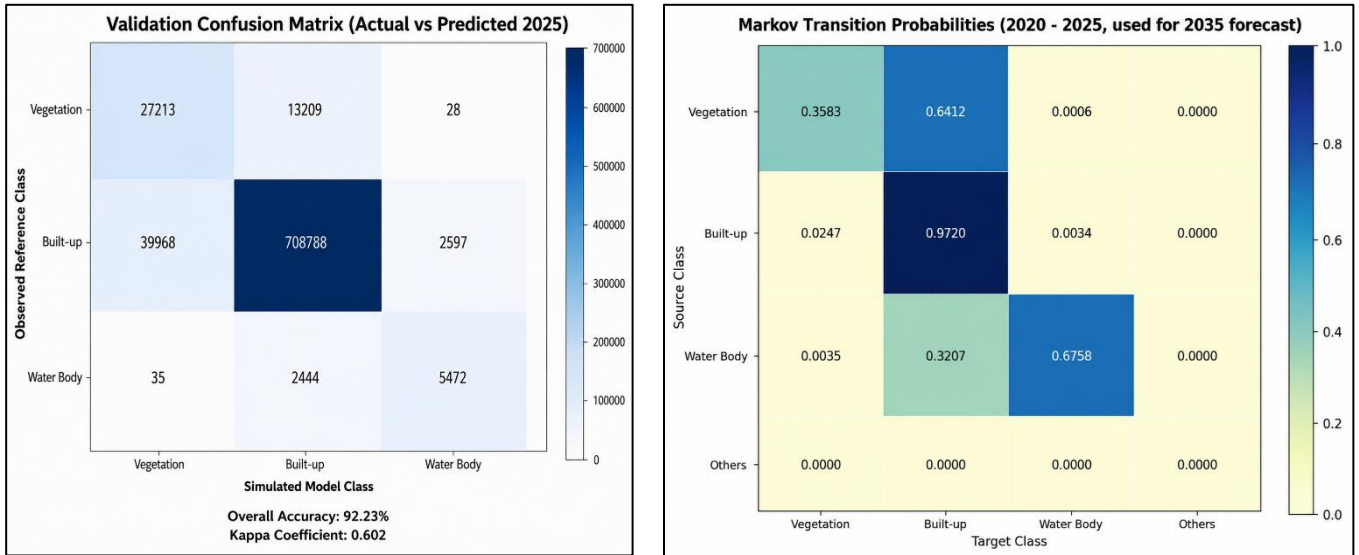


Fig. 7: (a) Confusion Matrix (CA-Markov), (b) Transition Probability Matrix (2025-2035)

5.2.4 Groundwater Depth Forecast using Ordinary Least Squares (OLS)

Groundwater depth in mbgl (metres below ground level) for the year 2035 has been predicted with the help of historical groundwater depth datasets of years 2012, 2016 and 2024, sourced from BWSSB, using Ordinary Least Square (OLS) model that has assumed a linear relationship between time (independent variable) and groundwater depth (dependent variable).

The OLS model equation is expressed as:

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

where:

- Y = Predicted groundwater depth (m below ground level),
- X = Year,
- β_0 = Intercept,
- β_1 = Regression coefficient (rate of groundwater change),
- ε = Random error term.

The regression coefficients were estimated by minimizing the sum of squared residuals:

$$\min \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

where:

- Y_i = Observed groundwater depth,
- $\hat{Y}_i$ = Predicted groundwater depth,
- n = Number of observations.

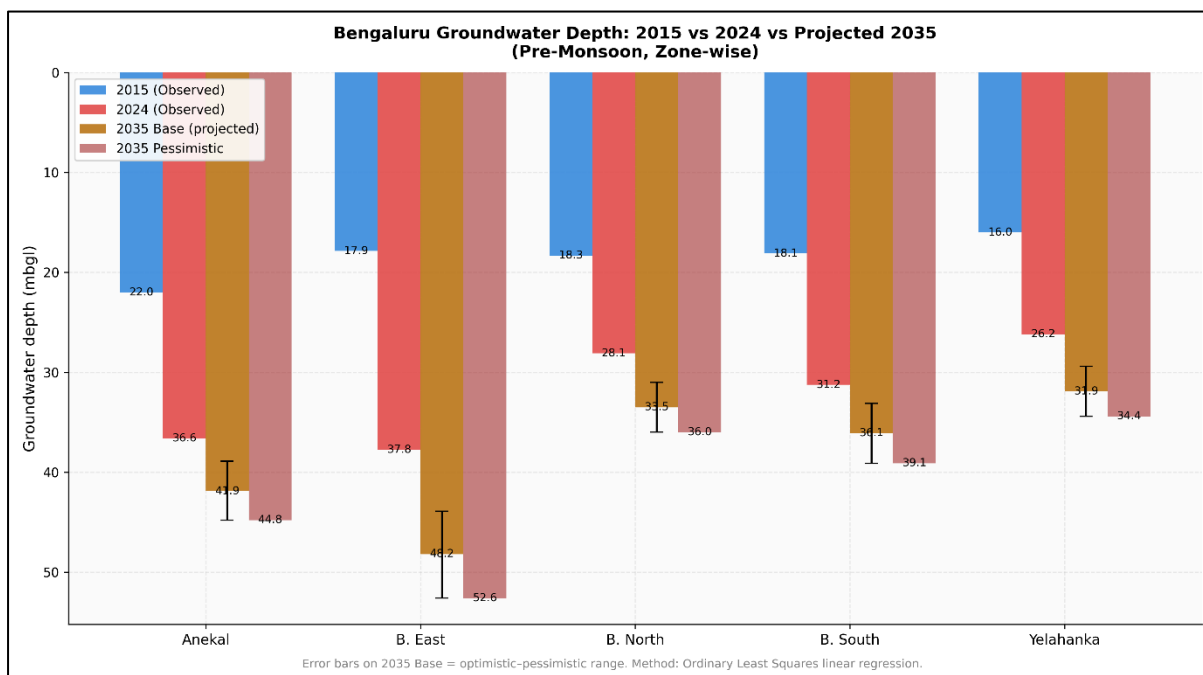
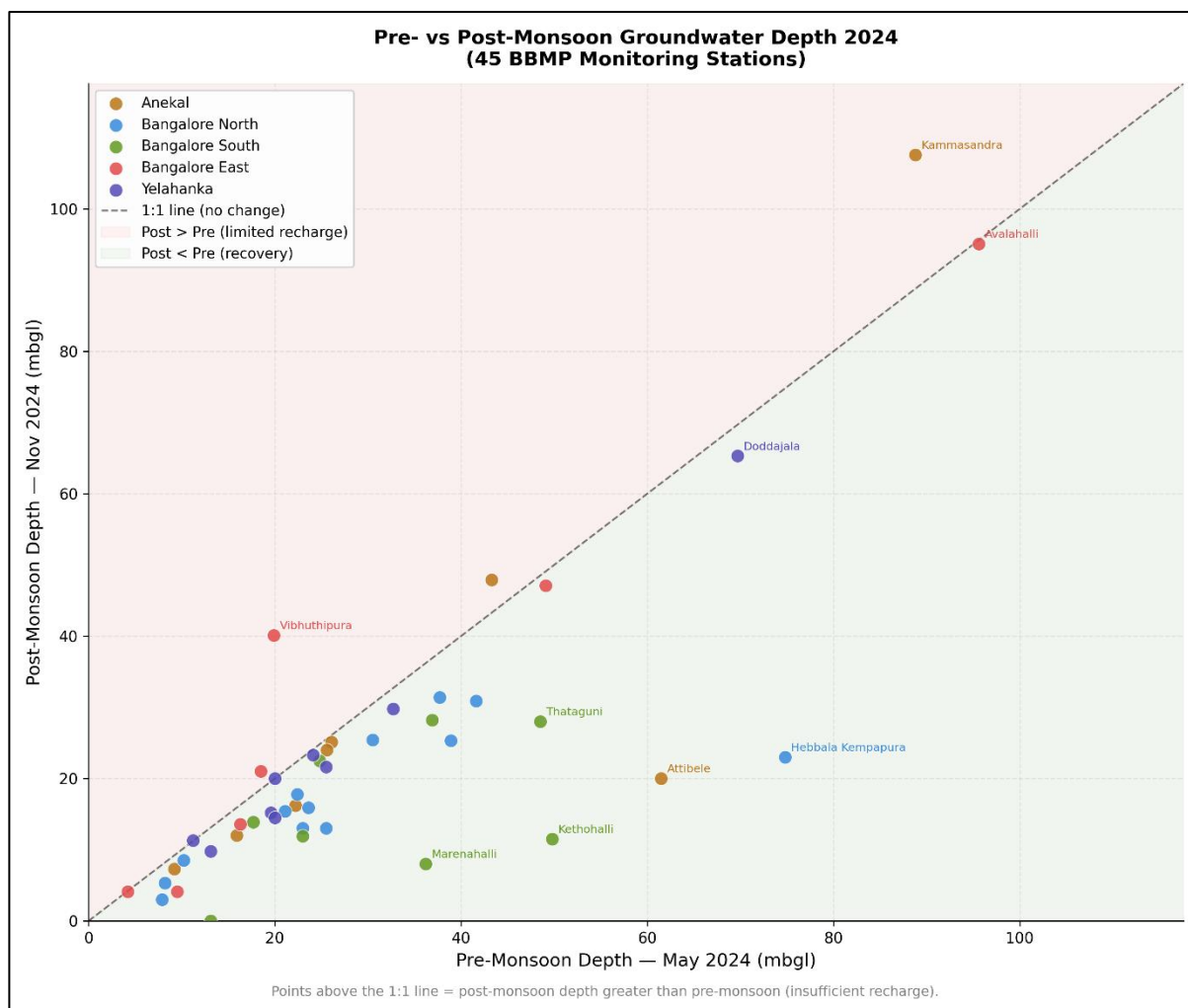


Fig. 8: (a) Pre v/s Post Monsoon Depth at certain locations, (b) Ward-wise 2025 v/s forecasted 2035 groundwater depth (in mbgl)

5.3 GIS-Based Spatial Analysis

5.3.1 Euclidean Distance Analysis

Euclidean Distance Analysis has been performed using ArcGIS Pro tool, to quantify the accessibility of the following:

- Distance to water pumping stations,
- Distance to existing water pipeline infrastructure,
- Distance to surface waterbodies.

The generated raster denotes spatial accessibility, where lower values indicate greater accessibility and higher values indicate poor accessibility.

5.3.2 Kernel Density Estimation (KDE)

Borewell density raster has been generated by performing a Kernel Density Estimation (KDE) technique that converts discrete borewell locations into a continuous density surface. KDE has been extensively performed in ArcGIS Pro,

5.3.3 Built-Up Density

Built-Up density maps for the years 2000, 2005, 2015, 2020 and 2025 have been generated from the Land Use Land Cover (LULC) maps for each year, respectively. Built-Up density analysis showcases the spatial variations in the degree of urbanisation within the BBMP region. Pixel-classified built up land were extracted and converted into a continuous density raster, using neighbourhood analysis in ArcGIS Pro.

5.3.4 Soil Infiltration Index (SII)

The Soil Infiltration Index raster has been prepared with the help of Google Earth Engine (GEE) using SoilGrids dataset. The index was estimated from the relative proportions of sand content and clay content, as soils containing high amounts of sand generally exhibits greater infiltration capacity. The derived index is normalised between 0 and 1. It is computed as follows:

$$SII = \text{Sand} / (\text{Sand} + \text{Clay})$$

5.3.5 Slope Derivation

Slope has been derived from the 30 m Shuttle Radar Topography Mission (SRTM) Digital Elevation Model (DEM). Slope represents the maximum rate of elevation change that influences surface runoff, infiltration and groundwater recharge potential. This has been performed extensively in ArcGIS Pro.

5.4 AHP-Based Generation of Training Dataset

5.4.1 Unavailability of Government-validated data

An Analytical Hierarchy Process (AHP) technique has been utilised through Multi-Criteria Decision Analysis (MCDA) for generation of training samples which include: 500 training points for three classes, “Low Stress”, “Moderate Stress”, “High Stress”. This has been performed with the help of 11 relevant hydrological, environmental and socio-economic variables for years 2025 and 2035.

Through extensive literature analysis and analysis of available government data, no officially labelled Urban Water Stress dataset currently exists for the city of Bengaluru. BWSSB Board has groundwater depth dataset ward-wise which has often been used as agency-published data when Bengaluru water crisis is researched. However, long-term water stress of Bengaluru, although is albeit mostly driven by groundwater, is a complex phenomenon which involve the interaction of different variables.

In view of this limitation, MCDA has been used for training labels generation, as formalised by Saaty (1980).

5.4.2 Predictor Variables

The 11 predictor variables used are:

Table 4: AHP Predictor Variables with Importance

Rank	Variable	Importance
1	Groundwater Depth	Very High
2	Borewell Density	Very High
3	Distance to Pumps	High
4	Built-Up Density	High
5	Population	High
6	Distance to Waterbodies	Moderate
7	Distance to Water Pipes	Moderate
8	Soil Infiltration Index	Moderate
9	Slope	Low
10	LST	Low
11	SPI	Low

5.4.3 Pairwise Comparison and Weight Assignment

Relative weights for the eleven predictor variables were derived through pairwise comparison in accordance with Saaty's nine-point fundamental scale, where each pair of criteria is judged in terms of its relative contribution to overall urban water stress and assigned an intensity value ranging from 1 (equal importance) to 9 (extreme importance).

5.4.4 Formulation of AHP

Priority weight vector has been subsequently derived by computing the principal eigenvector of the comparison matrix as:

$$A \cdot w = \lambda_{\max} \cdot w$$

Where, “w” is the eigenvector containing normalised priority weights of the eleven criteria and “ $\lambda_{\max}$ ” is the principal eigenvalue of matrix.

The maximum eigenvalue was computed as:

$$\lambda_{\max} = (1/n) \times \sum [(A \cdot w)_i / w_i]$$

Where, “ $(A \cdot w)_i$ ” is the i-th element of vector obtained by multiplying the comparison matrix by the weight vector and “ w_i ” is the corresponding i-th priority weight.

Then, Consistency Index (CI) is calculated as:

$$CI = (\lambda_{\max} - n) / (n - 1)$$

A Consistency Ratio (CR) was then formed by normalising the Consistency Index against the Random Index (RI) which consist of benchmark values corresponding to average CI expected from randomly generated reciprocal matrices.

$$CR = CI / RI$$

According to Saaty (1980), a CR below 0.10 indicates an acceptable level of coherence in the pairwise comparisons. This means, the judgement is accepted for further work. Higher than 0.10 CR means, judgement needs to be reconstructed and revised.

In this study, an acceptable Consistency Ratio (CR) of 0.05 has been achieved. And the AHP values are provided as such.

The AHP-weight table is as follows:

Table 5: Final AHP Weights of the Eleven Predictor Variables

Rank	Predictor Variable	AHP Weight	Weight (%)
1	Groundwater Depth	0.1837	18.37
2	Borewell Density	0.1633	16.33
3	Distance to Pumps	0.1224	12.24
4	Built-Up Density	0.1020	10.20
5	Population	0.1020	10.20
6	Distance to Waterbodies	0.0816	8.16
7	Distance to Water Pipes	0.0816	8.16
8	Soil Infiltration Index	0.0612	6.12
9	Slope	0.0408	4.08
10	LST	0.0408	4.08
11	SPI	0.0204	2.04

5.5 ML-Based Water Stress Prediction for 2025 and 2035

Two ensemble tree-based machine learning classifiers have been used for the final urban water stress prediction and hotspot mapping of Bengaluru for 2025 and forecasted 2035: Random Forest and XGBoost.

Random Forest and XGBoost have often been used in existing remote sensing and GeoAI literature for land cover classification, hazard susceptibility mapping and environmental indicators prediction. They are well-equipped to deal with high-dimensional, diverse, heterogeneous datasets and provide high model accuracy. Furthermore, they can handle the non-linearity among multi-source geospatial variables. This makes both the models ideal for this urban water stress project.

Variables such as Slope, Soil Infiltration Index, Distance to Pumping Stations, Distance to Water Infrastructure and Borewell Density have been kept consistent for both 2025 and 2035. However, remaining variables like Standardized Precipitation Index (SPI), Land Surface Temperature (LST), Built-Up Density, Population Density, Groundwater Depth and Distance to Waterbodies have been forecasted for the prediction year 2035.

5.1 Random Forest (RF) and XGBoost

Random Forest equation is expressed as:

$$\hat{y} = \text{mode} \{T_1(x), T_2(x), \dots, T_n(x)\}$$

In this equation, ‘ $\hat{y}$ ’ denotes the predicted class label for an input feature vector ‘ x ’, ‘ $T_i(x)$ ’ represents the class prediction generated by the i -th individual decision tree within the forest, ‘ n ’ is the total number of trees in the ensemble and the mode operator selects the class label that receives the greatest number of votes across all ‘ n ’ trees.

XGBoost Model have two equations: Objective Function –

$$\text{Obj}(\theta) = \sum_i l(y_i, \hat{y}_i) + \sum_k \Omega(f_k)$$

Here, ‘ $l(y_i, \hat{y}_i)$ ’ is the training loss function, typically the multi-class logarithmic loss for a three-class problem, which quantifies the discrepancy between the true label ‘ y_i ’ and the predicted label ‘ $\hat{y}_i$ ’ for the i -th training sample. ‘ $\Omega(f_k)$ ’ is a regularization term applied to the k -th tree ‘ f_k ’ in the ensemble and the summations are taken respectively over all training samples ‘ i ’ and over all ‘ k ’ trees added to the model.

Regularisation Function –

$$\Omega(f) = \gamma T + \frac{1}{2} \lambda \|w\|^2$$

‘ T ’ denotes the number of leaf nodes in a given tree, ‘ w ’ represents the vector of leaf weights and ‘ γ ’ and ‘ λ ’ are penalty coefficients that respectively control the cost of adding additional leaves and the magnitude of the leaf weights.

500 points for each class: “Low Stress”, “Moderate Stress” and “High Stress” were generated from the AHP-based Water Stress Map for both the years 2025 and 2035. The data is then split into a 70:30 ratio, where 70% of the labelled samples have been used for training and 30% were used for validation and testing.

Table 6: Performance Metrics of Trained Classifiers

Model	Year	Overall Accuracy (%)	Precision	F1-Score	Kappa
RF	2025	86.89	0.871	0.870	0.8033
XGBoost	2025	87.56	0.876	0.876	0.8133
RF	2035	85.08	0.849	0.848	0.7762
XGBoost	2035	87.97	0.880	0.879	0.8196

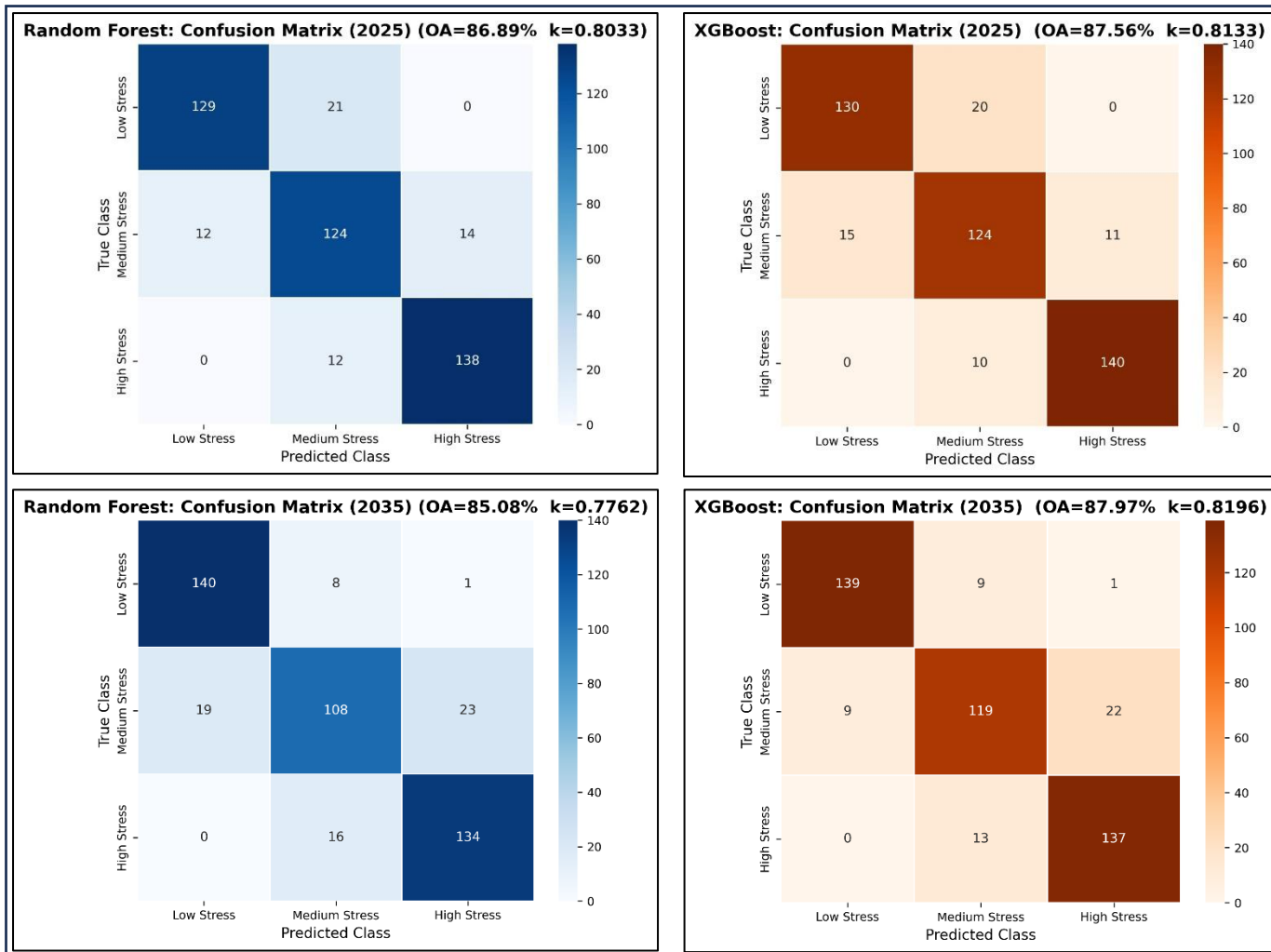


Fig. 9: Confusion Matrix of Random Forest and XGBoost Model (2025 & 2035)

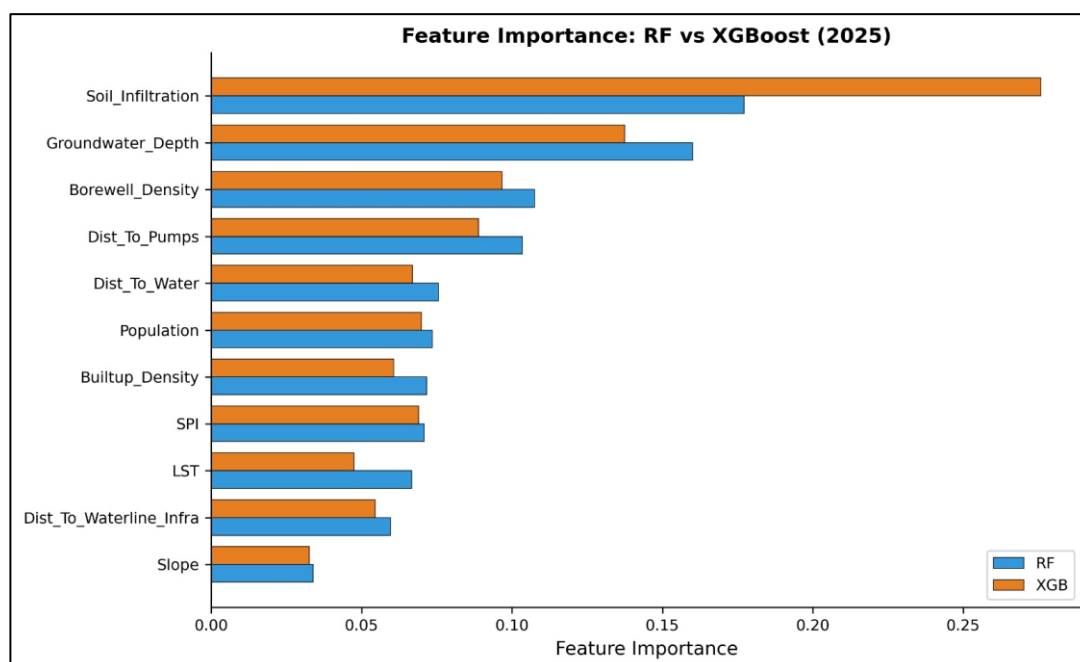


Fig. 10: Feature Importance Bar Chart (2025)

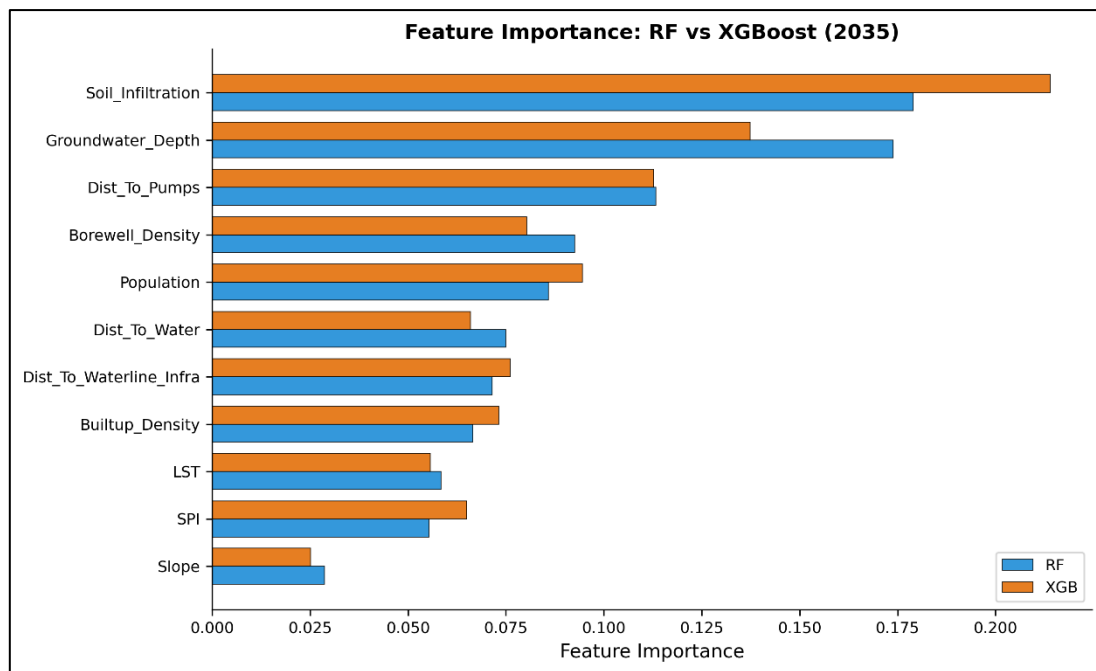


Fig. 11: Feature Importance Bar Chart (2035)

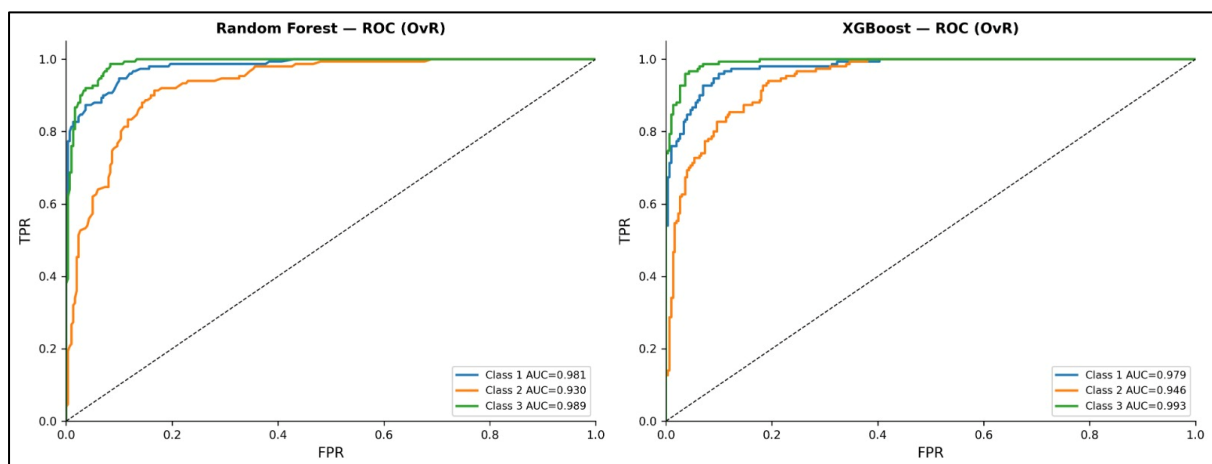


Fig. 12: ROC Curves of Random Forest and XGBoost Models (2025)

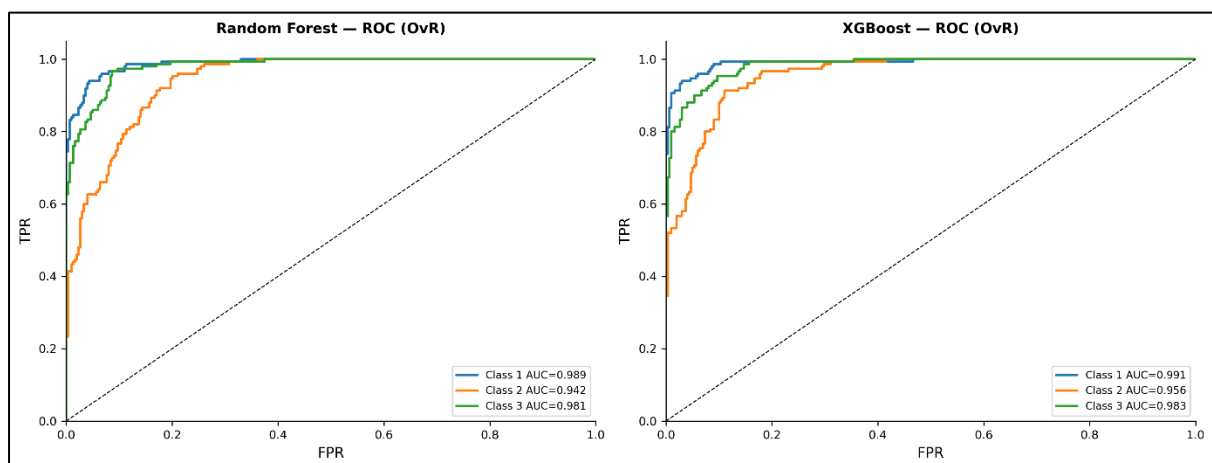


Fig. 13: ROC Curves of Random Forest and XGBoost Models (2035)

5.6 K-Means Clustering for Infrastructure Allocation

K-Means Clustering is an unsupervised classification algorithm that has been employed to segregate and classify high-priority pixels into three decentralised water infrastructure clusters:

- Recharge Wells
- Community Water Storage Tanks
- Rainwater Harvesting Structures

5.6.1 Input Variables for Clustering

The clustering analysis is performed using nine variables, of the 2035 dataset, encompassing the composite Urban Water Stress Index alongside underlying hydro-environmental and socio-economic predictors that determine the physical suitability of a given location for a particular category of decentralised infrastructure. The variables used are:

- Urban Water Stress Index
- Groundwater Depth
- Population Density
- Built-Up Density
- SPI
- LST
- Soil Infiltration Index
- Distance to surface waterbodies
- Borewell Density

5.6.2 Selection of Optimal Number of Clusters

Three independent cluster validation techniques have been employed to select the optimal number of clusters. The final number of clusters has been decided as three.

Elbow Method:

Elbow Method evaluates the Within-Cluster Sum of Squares (WCSS), which quantifies the total squared distance between each observation and its assigned cluster centroid across the entire dataset:

$$WCSS = \sum_k \sum \|x_i - c_k\|^2, x_i \in \text{Cluster } k$$

Silhouette Score:

Silhouette Score provides per-observation measure of clustering quality by jointly evaluating intra-cluster cohesion and inter-cluster separation for each individual observation:

$$s(i) = (b(i) - a(i)) / \max \{a(i), b(i)\}$$

where:

- $a(i)$ = intra-cluster distance,
- $b(i)$ = mean nearest-cluster distance.

$s(i)$ value ranges between -1 to +1. Values closer to +1 means that observation is well matched to its own clusters and poorly matched with neighbouring clusters.

Davies-Bouldin Index (DBI):

Davies-Bouldin Index is a further validation criterion based on the ratio of within-cluster scatter to between-cluster separation, computed as:

$$DBI = (1/K) \sum_k \max_{j \neq k} \{(\sigma_k + \sigma_j) / d(c_k, c_j)\}$$

where ' σ_k ' and ' σ_j ' denote the average distance between each observation in clusters 'k' and 'j' respectively and their corresponding centroids. ' $d(c_k, c_j)$ ' denotes the Euclidean distance between the centroids of clusters 'k' and 'j' and the maximum operator identifies, for each cluster 'k', the neighbouring cluster 'j' that yields the least favourable (highest) scatter-to-separation ratio.

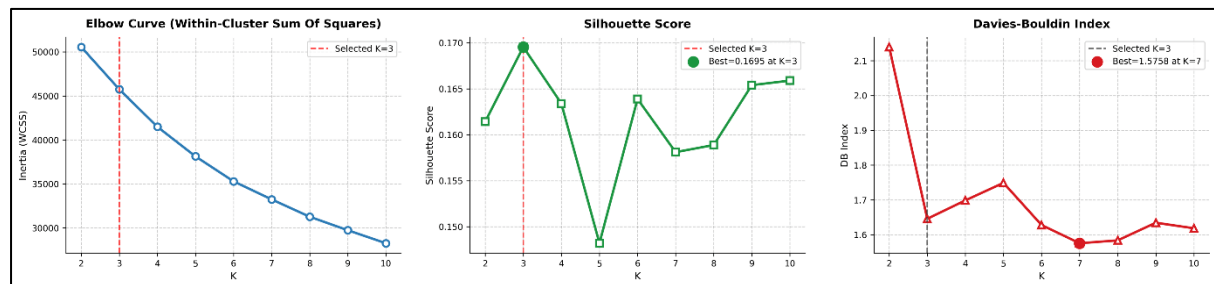


Fig. 14: Optimal Number of Clusters Selection Methods

5.6.3 Infrastructure Allocation Based on Cluster Characteristics

Cluster 1: Greater groundwater depth, higher borewell density and favourable soil infiltration capacity (Assigned to Recharge Wells).

Cluster 2: High population and built-up density, high water stress, less soil infiltration capacity (Assigned to Community Water Storage Tanks).

Cluster 3: Higher distance to surface water bodies, high SPI, moderate soil infiltration capacity (Assigned to Rainwater Harvesting Structures).

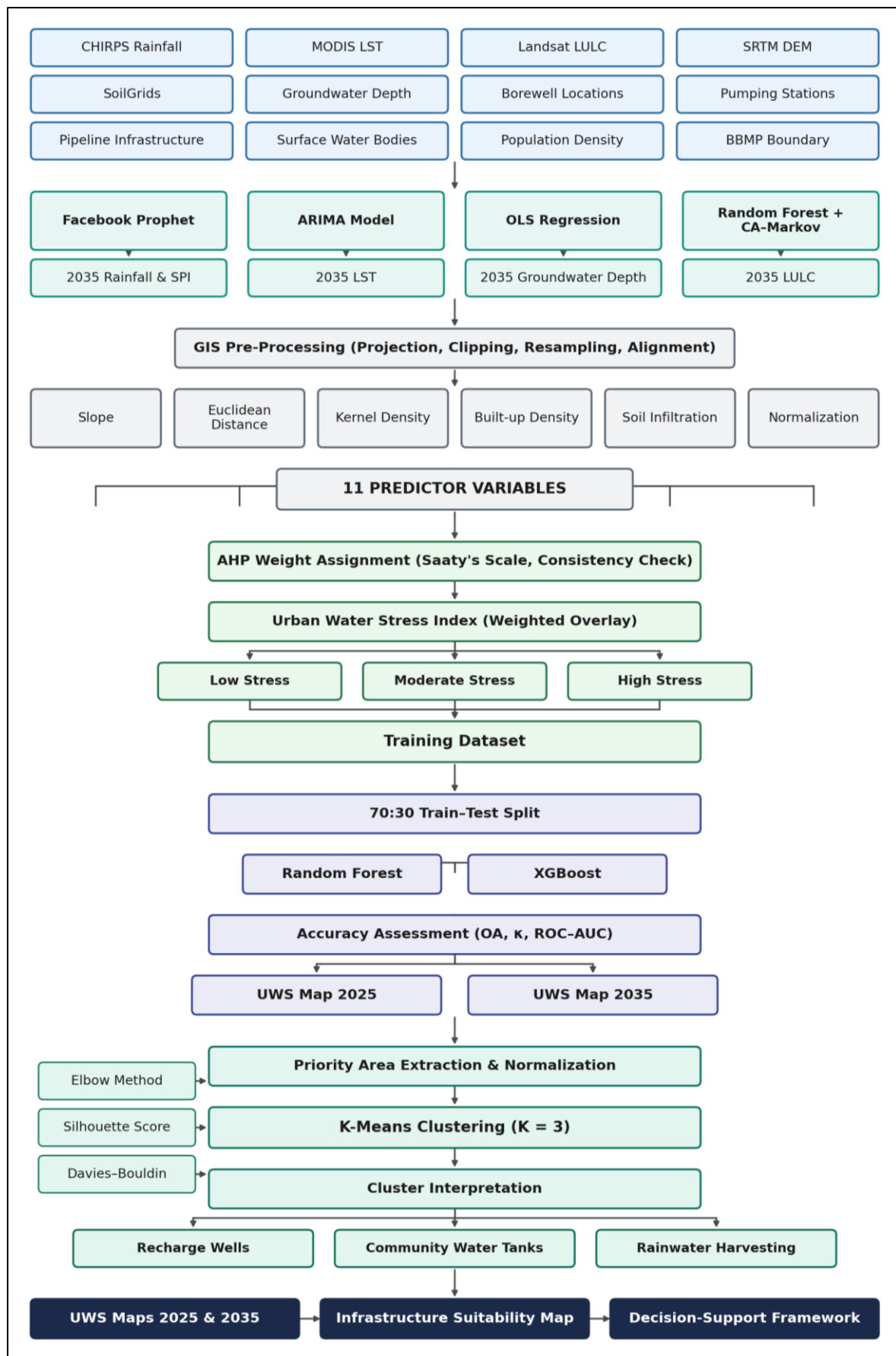


Fig. 15: Methodology Flowchart

CHAPTER 6:

Results & Discussions

6.1 Spatial Distribution of Variables (2025 v/s 2035)

6.1.1 Rainfall and Standardized Precipitation Index (SPI)

Rainfall analysis (Figure 16) of Bengaluru has largely revealed a distinct north-east to south-west gradient for the year 2025. Annual rainfall has ranged from 1022 mm in the north-central region to as high as 1247 mm over south and south-western periphery. This rainfall pattern is particularly significant since it is consistent with Bengaluru's peninsular plateau geology. Therefore, it can be established that Bengaluru's elevated western and southern fringes receive significantly higher rainfall than the rest of the city. Prophet-based 2035 projection retains the same broad spatial structure but exhibits a systematic redistribution. Higher rainfall is predicted to occur further towards the south-west region, while a larger portion of east and north-eastern regions are projected to receive comparatively lower rainfall amounts.

The SPI-12 distribution for 2025 (Figure 16) highlights the north-central and north-eastern wards, together with a narrow strip along the eastern Bengaluru, registering Near Normal to Slightly Wet conditions, while the western, south-western and southern fringes fall predominantly in the Moderately Wet to Highly Wet categories. However, by 2035, the Near Normal class expands markedly to cover almost the entire northern half of Bengaluru and the transition to wetter categories becomes increasingly confined to the southeastern quadrant. It has been further observed that the wards falling within the expanding Near Normal class are largely coincident with the densely built-up core identified during Built-Up Density analysis. This observation confirms the bias that long-term below-mean rainfall may have caused these particular areas to become the emerging drought-susceptible zones of the city.

However, for both 2025 and 2035, spatial variability of rainfall and SPI are not particularly large enough to be considered a major driver of Bengaluru's urban water stress. Rainfall and subsequently, SPI are therefore considered as auxiliary variables that influences overall groundwater recharge and distribution, rather than direct drivers of water stress.

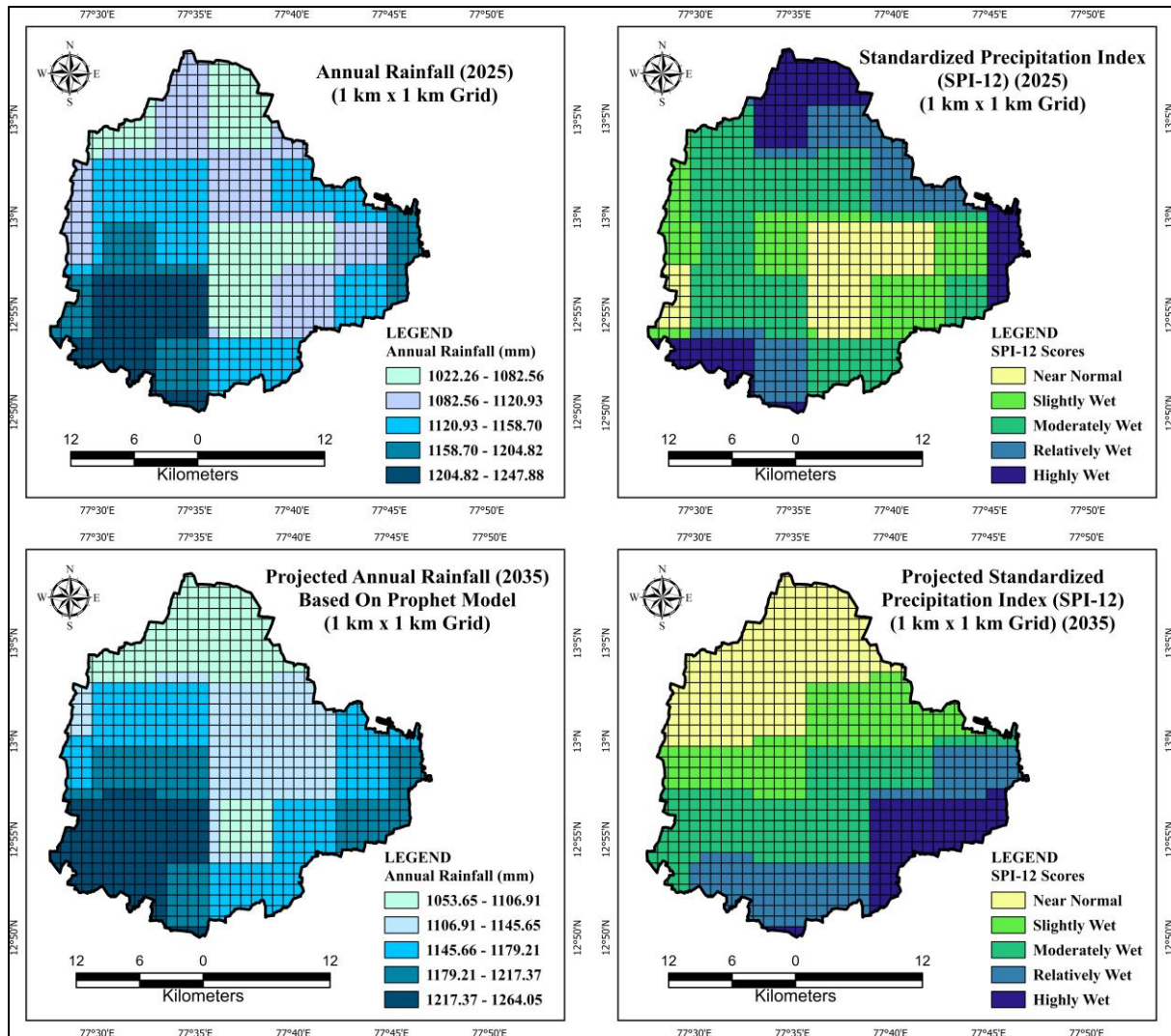


Fig. 16: Spatial Distribution of Annual Rainfall (mm) and SPI-12 for 2025, (b) Prophet-Based Projections for 2035 (At 1 km × 1 km Grid Resolution).

6.1.2 Land Surface Temperature (LST)

During 2025, a noticeably pronounced Urban Heat Island (UHI) effect has been observed in western and north-central zones of BBMP (Figure 17), where daytime temperatures have reached up to 30.2°C, which is in sharp contrast to residual cooler pockets having temperatures of 25.0–26.8°C associated with the Bellandur–Varthur wetland and lake complex in the southeast. The spatial coincidence of the hottest LST zone with the oldest and most consolidated built-up core is consistent with the classical UHI mechanism, whereby the low albedo, high thermal admittance and reduced evapotranspiration of concrete, asphalt and roofing materials elevate surface temperatures relative to vegetated or water-covered land.

The 2035 ARIMA-predicted LST (Figure 17) portrays an intensified alarming scenario in which the highest LST dominant class is expected to rise up to 33.6–34.3°C, which is a rise of about 6°C in just a decade. Observations show that extreme hotspots are expected to extend further towards the western wards, as well as encroach upon Bengaluru’s northern periphery. South-eastern wetlands and lake pockets are expected to continue having daytime temperatures several degrees below the city’s average.

Such alarming findings of this project is directionally consistent with independent studies of Bengaluru's thermal environment that have linked the loss of vegetation and water spread area to statistically significant LST increases. Higher LST suggests a increase in the city’s per capita domestic and institutional water consumption as well as higher irrigation demands. Coupling this with accelerated urbanisation of Bengaluru, the dire situation of Bengaluru’s acute water stress with water demands highly surpassing available water supply is naturally expected. LST is therefore considered as a factor of moderate influence to Bengaluru’s water supply crisis scenario.

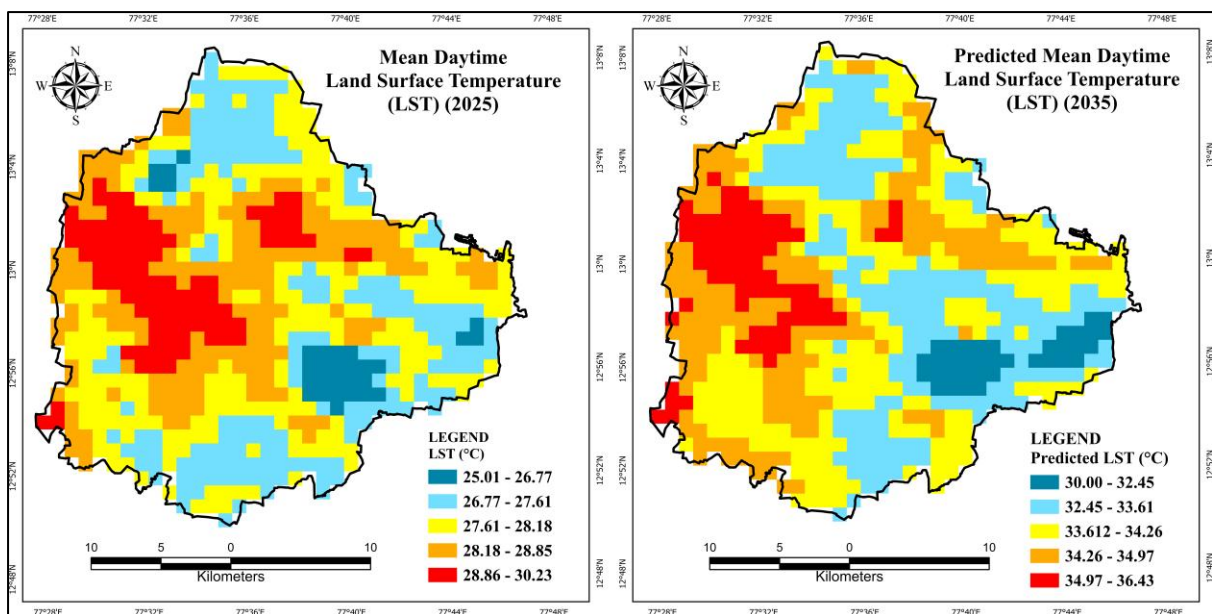


Fig. 17: Daytime LST (°C) for 2025 and ARIMA-Predicted Daytime LST (°C) for 2035.

6.1.3 Land Use & Land Cover (LULC)

The supervised Random Forest-based multi-year LULC classification of Bengaluru (Figure 19) documents one of the most rapid urban land conversions reported for any Indian metropolitan region. Built-up area within the 715.81 km² BBMP boundary increased from 521.31 km² (72.8%) in 2000 to 688.25 km² (96.2%) in 2025, while vegetation cover collapsed from 177.52 km² (24.8%) to just 21.32 km² (3.0%) over the same period and waterbody extent nearly halved

from 10.36 km² to 6.24 km². The steepest built-up gain has been observed between 2000 and 2005 (96.05 km² in five years). This alarming increase coincides with Bengaluru's IT-led economic expansion and subsequent peripheral annexation of villages into the city's administrative limits during this period.

By 2035, built-up area is projected to reach 696.19 km² (97.3%), while vegetation is projected to decline further to 14.56 km² (2.0%) and waterbody area to contract to 5.06 km² (0.7%) (Figure 18). The residual vegetation patches by 2035 are confined almost entirely to the extreme northern, south-western and southern peripheral regions, mirroring the patterns of highest rainfall and lowest LST as documented in previous analysis. The almost near-total decline of remaining surface urban lakes and tank networks of Bengaluru is of primal concern as these waterbodies have historically served as groundwater recharge structures, freshwater alternatives and flood buffers. This further reinforces the well-known notion that such high percentages of built-up expansion and accelerated increase of impervious surfaces is one of the most important drivers of Bengaluru's urban water stress.

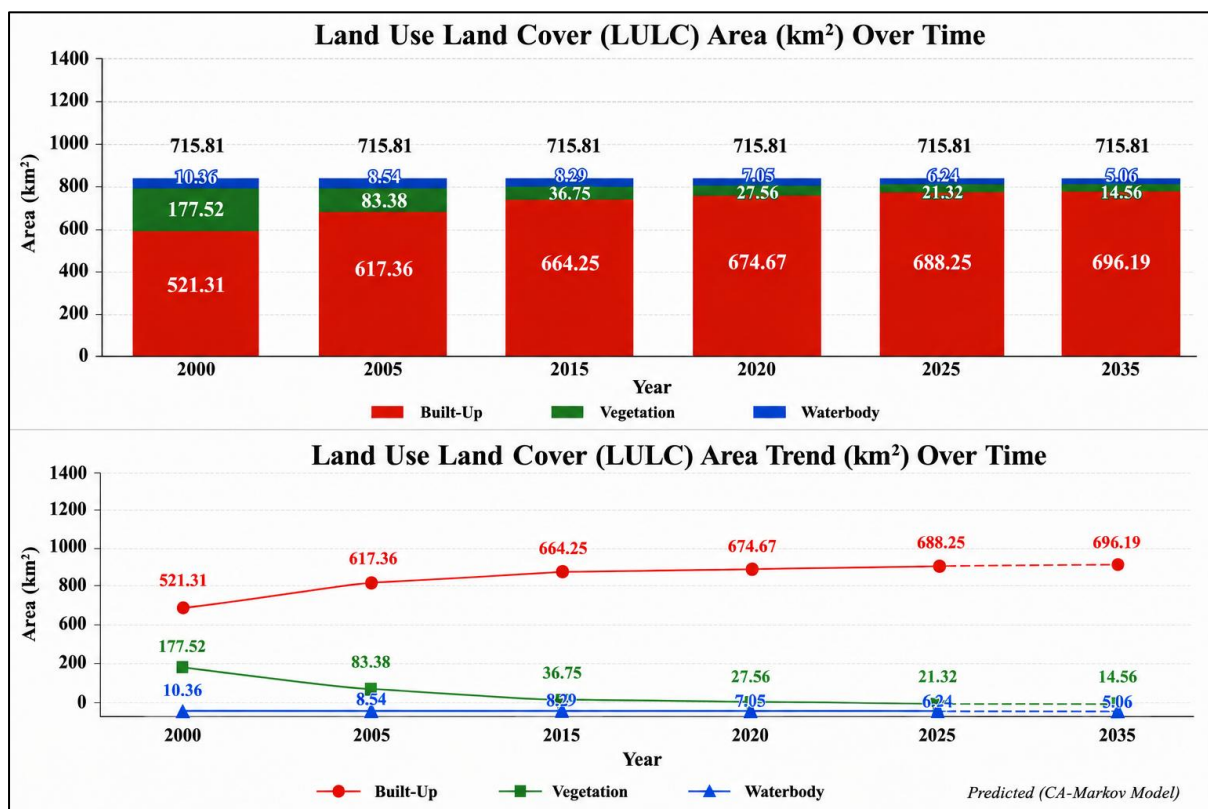


Fig. 18: Temporal Trends in Built-Up, Vegetation and Waterbody Area (km²) (2000-2035)

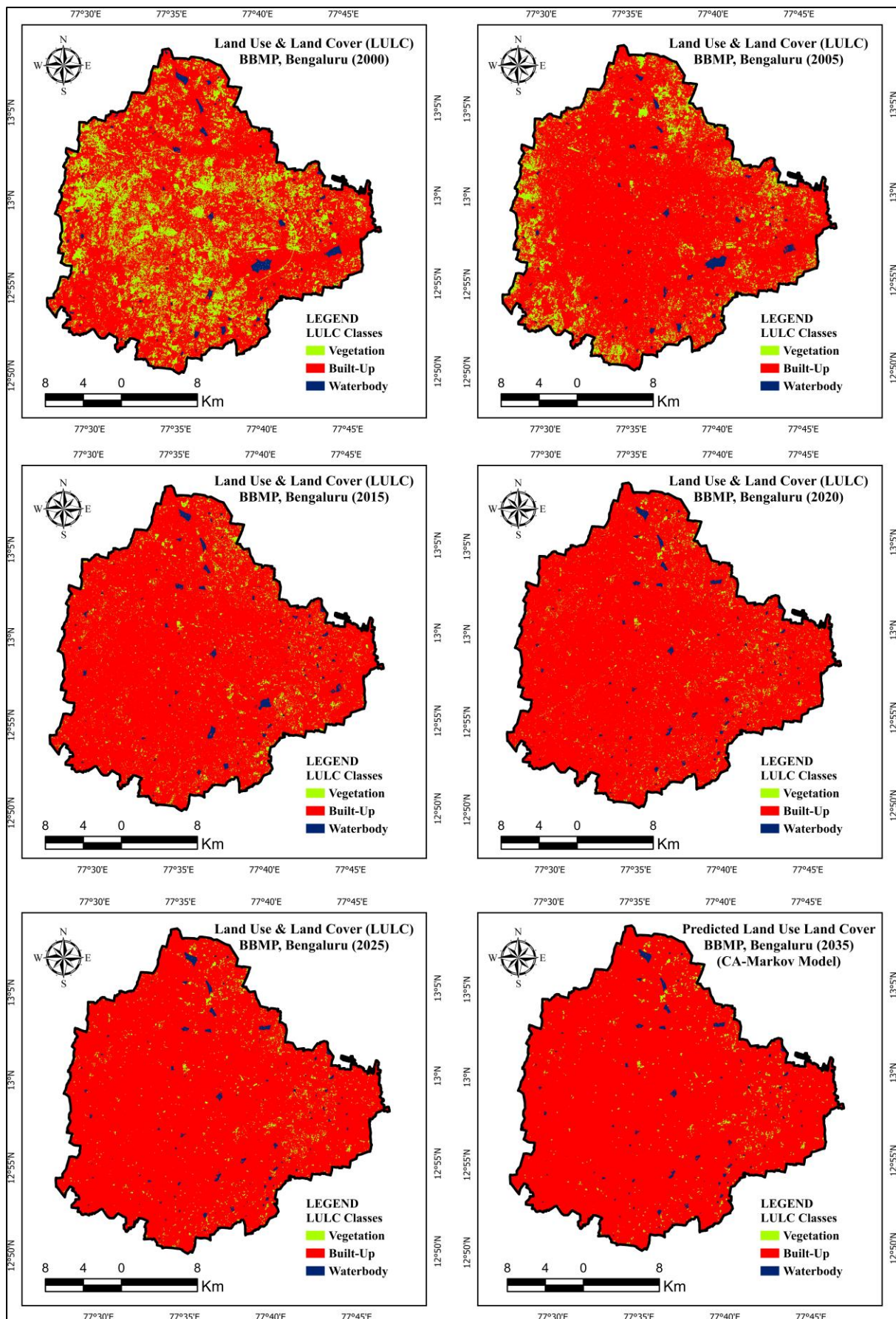


Fig. 19: Land Use & Land Cover (LULC) Classification (2000-2035)

6.1.4 Groundwater Depth, Borewell Density, Pumping Stations and Existing Water Pipeline Infrastructure

The 2025 interpolated groundwater depth map (Figure 20) showcases depth-to-water-table values ranging from 4.2 mbgl in certain scattered pockets, especially along residual drainage lines or near existing urban lakes, to as high as 95 mbgl in western and south-western Bengaluru. This finding rightfully confirms that due to Bengaluru's hard rock (Peninsular Gneiss) aquifer system, groundwater is abundant only in discontinuous weathered and fractured zones rather than in a laterally continuous unconfined aquifer. The OLS-based 2035 prediction points towards a dire and alarming scenario, where the steepest projected groundwater level declines are concentrated in the same western and central wards that already exhibit the deepest 2025 levels as well as densest borewell concentration.

Borewell Density analysis has been kept consistent for 2025 as well as 2035, because it is difficult to accurately predict and map locations where new borewells are expected to be dug by 2035. Similar consistency is therefore kept for distance to pumping stations and distance to existing waterline infrastructure. By 2025, the distribution of existing borewells has reached a high of 157 borewells per km² in a broad cluster spanning the western and north-central wards whereas southern and eastern regions are marked by a lesser dependence on borewells. Such distinction is noticeable when we look at Bengaluru's historical settlement distribution, where western core has always comprised some of Bengaluru's oldest and densest regions with piped Cauvery water coverage lagging behind high population growth. As such, water supply for these areas is compensated through private and community borewell drillings, which in turn has further accelerated local aquifer depletion, causing a long-term urban water stress.

Distance to nearest water pumping station further highlights how far away certain eastern and south-eastern periphery wards are from direct access to water pumping stations. Particular regions of Bengaluru are located as far as 10 km away from the nearest water pumping station and therefore, these regions remain dependent on local and private borewells as water supply alternative. This also reflects the historically sparse and centrally-clustered layout of BWSSB's primary pumping infrastructure relative to the areally expansive Bengaluru's BBMP boundary.

Distance to existing waterline infrastructure may be considered another axillary driver of Bengaluru's water crisis but this has also been analysed to get a rough sketch of which parts of the city lie near existing governmental water infrastructure and pipelines, as these areas are

expected to be greatly benefited from government-based community water supply. Areas farther away from waterlines are less likely to be directly benefitted from governmental water supply and therefore, are more likely to be dependent on private borewells and surface waterbodies as water source alternatives. The distance to nearest mapped waterline is comparatively short across most of the city with 540 m as mean and 372 m as median distance. However, outlier exists within the city's north-western and south-eastern extremities, which include certain regions located more than 2-3 km away from existing waterlines.

Increasing levels of groundwater depth, higher frequency of borewell locations and larger distances from nearest pumping stations and existing water infrastructure are therefore rightfully considered some of the most important reasons for Bengaluru's current water crisis.

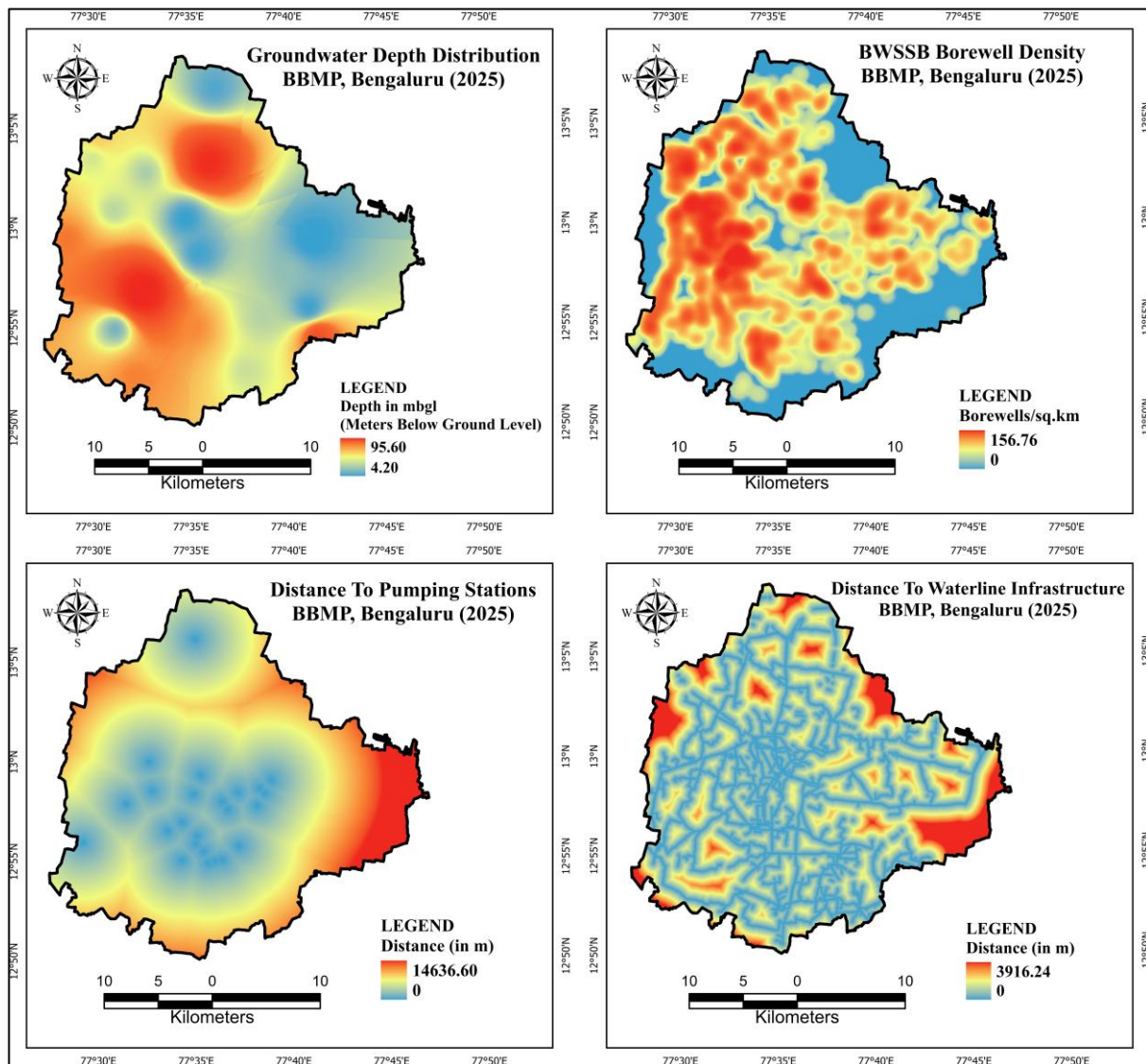


Fig. 20: Groundwater Depth (mbgl), Borewell Density (sq. km), Distance to Pumps (m) & Distance to Water Pipeline Infrastructure (m)

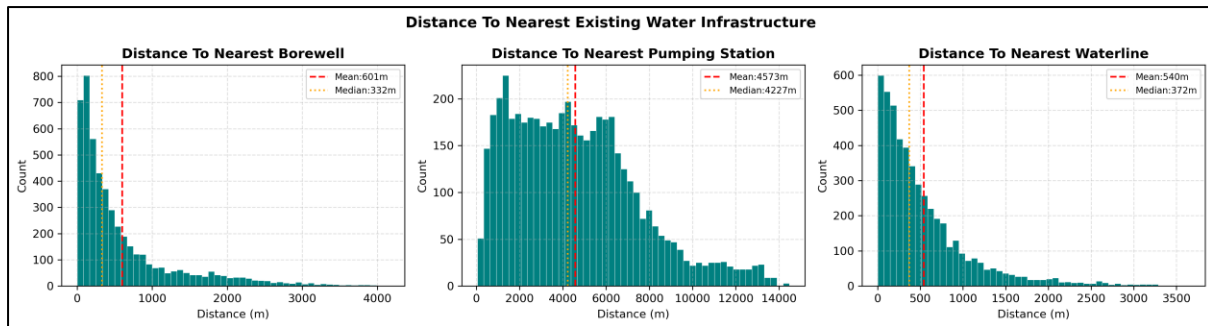


Fig. 21: Distances to Nearest Water Infrastructures (Borewells. Pumps. Water Pipelines)

6.1.5 Population & Built Density

The 2011 Census-based Bengaluru's ward-wise population map (Figure 22) shows that the highest population counts (41,028–43,243 persons per ward) are largely concentrated in a discontinuous ring surrounding the central business district and extending into the southern and north-eastern wards, with comparatively lower densities in the extreme northern and southwestern periphery. This map further confirms the Built-Up Density analysis conducted earlier which states that highest built density is observed in central BBMP as well as north-eastern areas of the city. For 2035, WorldPop Global Projection for 2035 data has been utilised which further highlights the acceleration of urbanisation and built-up intensification throughout the entire city.

Population Density analysis coupled with Built area delineation is of high importance in view of the notion that rapid urbanisation is one of the most important contributors of Bengaluru's water scarcity. Large-scale population growth negatively affects BWSSB's water supply capacity and potential since per-capita water demand scales directly with resident population and the spatial mismatch between the highest-population wards and the historically under-served western borewell-dependent belt is a key driver of this compound stress pattern.

6.1.6 Elevation & Slope

Slope derived from DEM has also been analysed as considered as an input variable for machine learning-based water stress prediction. While slope exerts a comparatively minor direct influence on water demand, it is hydrologically significant for infrastructure siting because steeper terrain increases surface runoff velocity and reduces the residence time available for infiltration, favouring the location of rainwater harvesting and check-dam-type structures on gentler slopes where ponding time and infiltration opportunity are maximised.

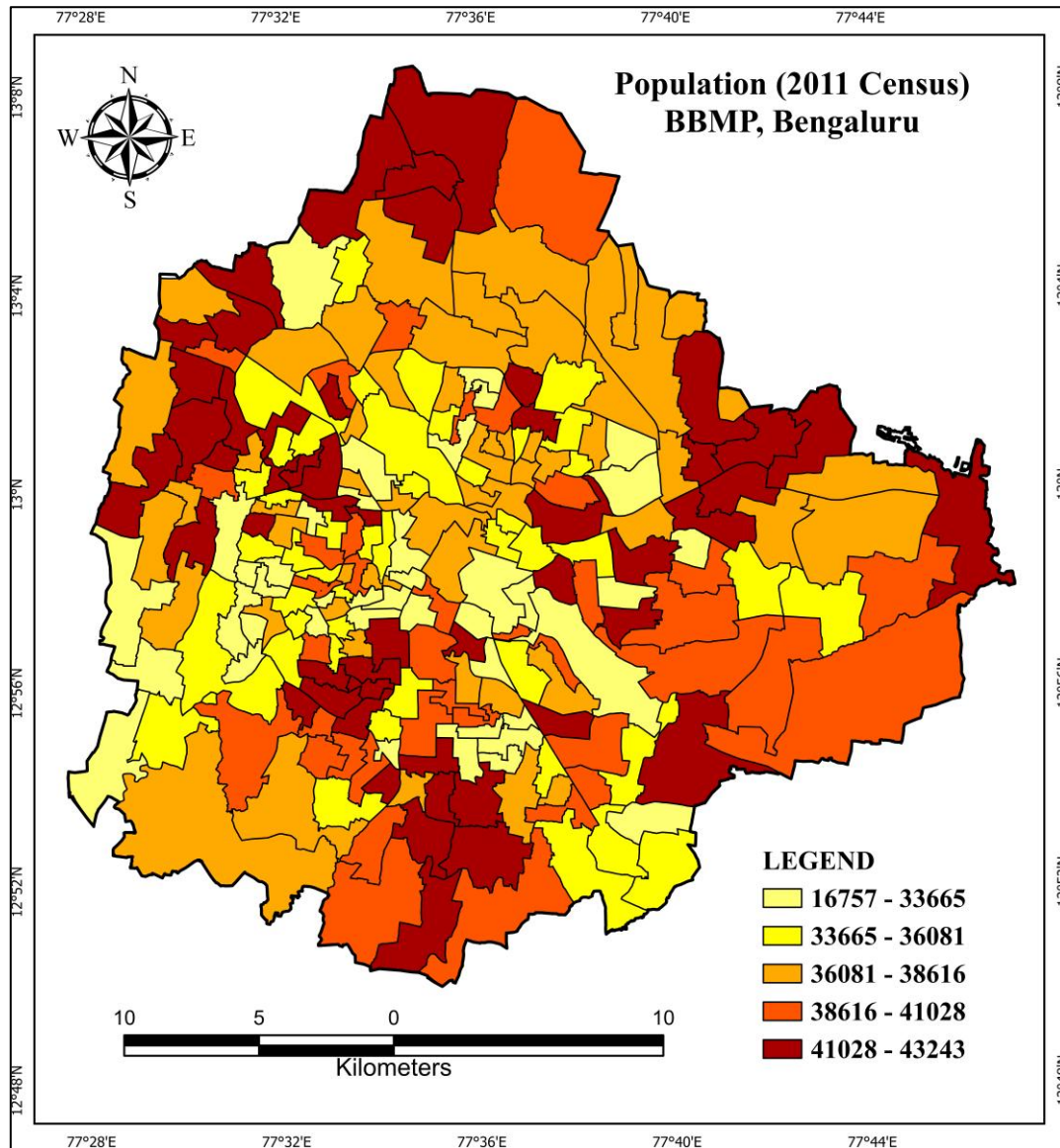


Fig. 22: Ward-Wise Population of Bengaluru (Based on Census of India, 2011)

6.1.7 Soil Infiltration Index

The normalised Soil Infiltration Index map (Figure 23) has been kept consistent for both 2025 and 2035 because soil type and patter of a city is not expected to go through major transformation over a ten-year period. Soil Infiltration Index visualisation shows that the Very Low Infiltration class is highly prominent across the central and parts of Bengaluru, consistent with the clay-rich lateritic and red soil associations typical of the Bengaluru plateau. High to Very High Infiltration is confined across discrete pockets along the western and southern boundary and a smaller cluster in the extreme north. Soil Infiltration Index is a moderate to high importance factor of Bengaluru's water stress because infiltration capacity of soil directly affects natural groundwater recharge potential. Therefore, this index is used both for machine-

learning based water stress prediction as well as decentralised infrastructure allocation-optimisation in later part of this research.

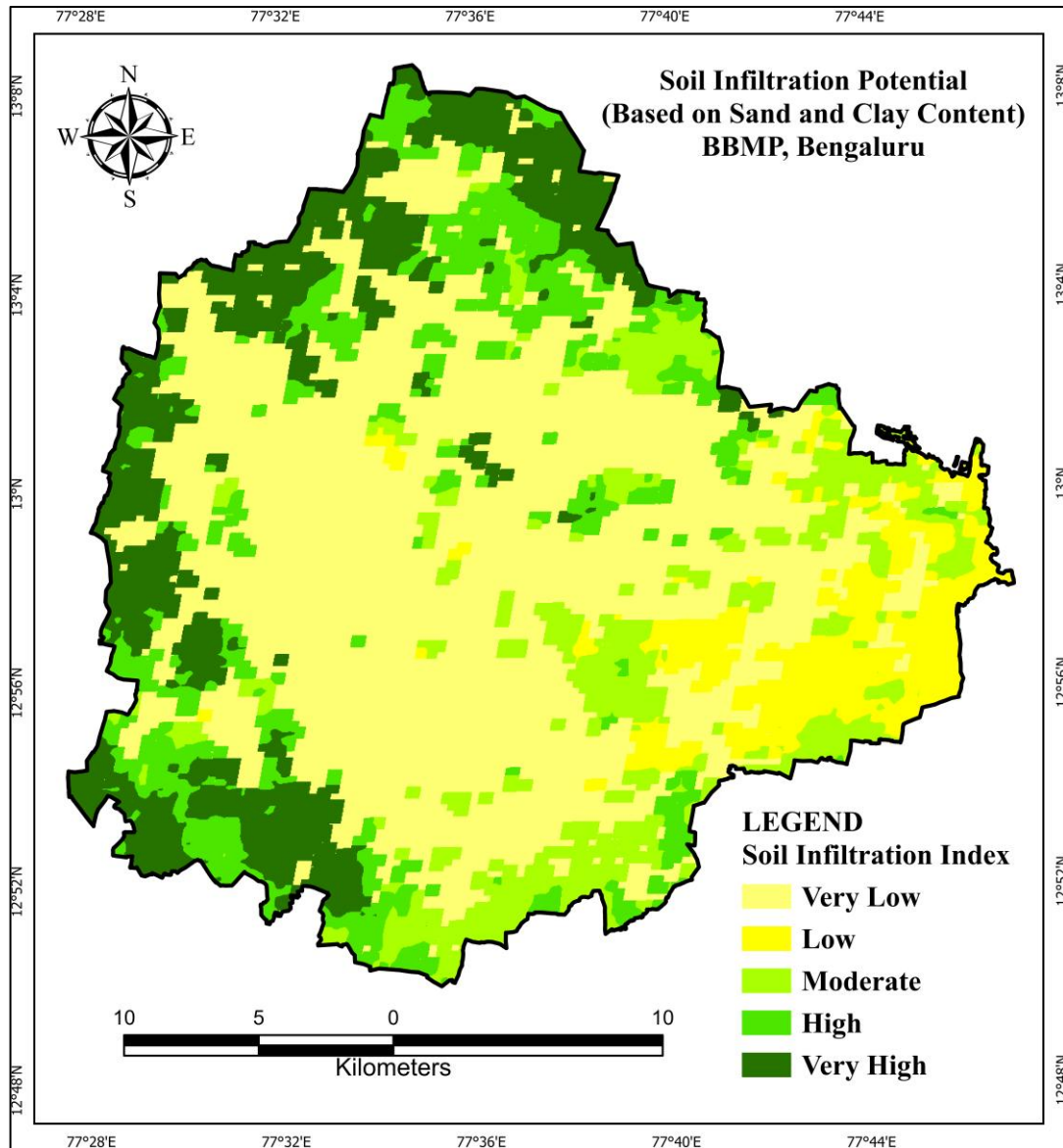


Fig. 23: Soil Infiltration Index of Bengaluru

6.1.8 Distance to Surface Waterbodies

As previously highlighted, there is a rapid acceleration of Built-Area in Bengaluru, coupled with encroachment over surrounding vegetation as well as surface lakes and wetlands. With large-scale groundwater depletion, urban lakes used to be an alternative to Bengaluru's water supply to some extent. However, it has been projected that, most urban lakes will be converted to built area by 2035. As such, distance to surface waterbodies has been considered as a factor of moderate importance for machine learning based urban water stress prediction of the city.

6.2 Random Forest and XGBoost-Based Water Stress Hotspots Prediction

The Urban Water Stress Index as well as hotspots prediction has been performed using AHP-based training labels, with the incorporation of eleven driver variables, out of which, four have been forecasted for the year 2035, using suitable appropriate forecasting models.

For the year 2025, both Random Forest and XGBoost models (Figure 24) have provided almost spatially consistent results with “High Stress” zones dominating large clusters in north-central and western wards, certain regions in the eastern and south-eastern Bengaluru and dispersed set of patches in some southern wards. These hotspots zones correspond highly to spatial overlap identified, between high borewell density, long distances to water pumping stations, dense population and elevated LST. XGBoost has produced a visibly more fragmented and finer-grained “High Stress” pattern than Random Forest, with a greater number of small, spatially discontinuous “High Stress” patches embedded within the “Moderate Stress” matrix, particularly in the central and southern wards. It is note-worthy that “Low Stress” class has been identified in major parts of northern Bengaluru and south-central regions, for both the models.

For the 2035-predicted Urban Water Stress maps (Figure 24), both the models show a pronounced expansion and coalescence of the “High Stress” zones relative to the 2025 baseline for both classifiers. The three discrete 2025 hotspot clusters largely merge into a single, near-contiguous “High Stress” corridor traversing the city from the north-central wards through the historic core and into the eastern boundary, while the areal extent of “Low Stress” contracts substantially and becomes increasingly confined to the extreme periphery. The intensification of this water stress can be attributed to the following reasons: continued LST warming, near-saturation built-up expansion with residual vegetation loss and progressive groundwater deepening and depletion of existing groundwater.

XGBoost has consistently out-performed Random Forest model for urban water stress prediction for both 2025 and 2035, achieving an Overall Accuracy and Kappa of 87.56% and 0.8133 for 2025 and 87.97% and 0.8196 for the year 2035, respectively. A Kappa above 0.80 suggests “Near Perfect Agreement”. Therefore, it can be concluded that both the models are internally consistent. XGBoost-based percentages of “Low”, “Moderate” and “High” stress classes percentages are as follows:

Table 7: Observed Stress Classes: Area (Hectares) and Percentage of Area (%)

Stress Level	Area (Ha): 2025	Area % (2025)	Area (Ha): 2035	Area % (2035)
Low Stress	232646.3	32.54	17288.2	24.37
Moderate Stress	382978.5	46.59	35682.8	50.31
High Stress	172494.2	20.87	17959.6	25.32

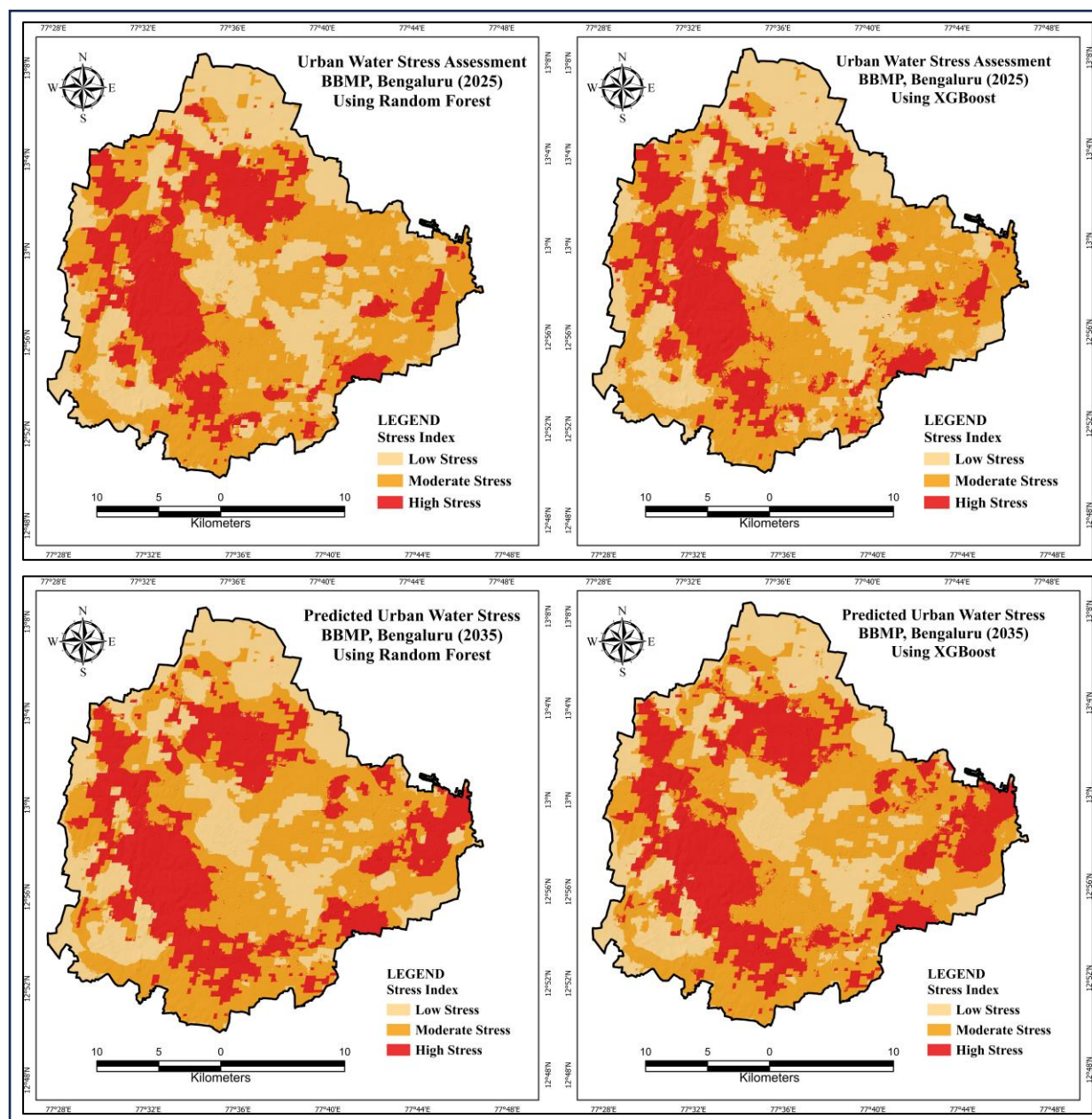


Fig. 24: Machine-Learning-based Urban Water Stress Prediction (2025 & 2035)

6.2.1 External Corroboration of 2025 and predicted-2035 map of Bengaluru's Water Stress

Both Random Forest and XGBoost based predictions and modelling has been validated in terms of an external government groundwater and related data-based corroboration. This has been performed especially due to this project's limitation of model development without real-life stress labels, and relying on an AHP-based framework for training samples generation.

Both Random Forest and XGBoost based 2025 "High Stress" clusters are concentrated in north-central, western and eastern BBMP wards. These spatially match with a 2025 BWSSB survey that identified 80 wards and 110 peripheral villages of BBMP which are facing acute groundwater shortage. Areas like Mahadevapura and Whitefield are facing some of the most severe water stress in Bengaluru, and these areas are correctly classified as "High Stress" by both the models.

An independently constructed ward-level Water Security Index built by WELLS Labs based on Cauvery-supply reliability, borewell counts and land-use data also report similar structural patterns. Both the models of this study correctly predicts that wards with dependable long-standing Cauvery water supply coverage ranks in "Low Stress" (especially northern Bengaluru wards) whereas eastern and central wards are historically the most under-served.

Similarly, the 2035 prediction showing an intensification and eastward spread of Bengaluru's water stress is expected to be accurate and can already be observed through real-time BWSSB as well as IISc reports. Bengaluru's groundwater is declining by 5 m annually whereas surrounding villages are witnessing declines of 10-15 m a year. This further highlights and confirms the findings that the water stress is not only intensifying but also gradually moving outwards of BBMP's main urban core.

A final peer-reviewed 2025 study in Water Policy by IWA Publishing that has used a data-driven approach that has combined supply-demand trends, financial data and climate projections to map the city's unsustainable water supply system, has also highlighted that central Bengaluru wards, especially those in Bengaluru-East, Bengaluru-South, Bengaluru-West and Mahadevapura are facing some of the worst water crisis in the city, whereas wards in Yelahanka, Bommanahalli, Dasarahalli lie in low to moderate water stress zones. This has also been appropriately spatially confirmed through this project's Random Forest and XGBoost models. Therefore, it can be inferred that the findings of this study are accurate and acceptable.

6.3 Optimal Decentralised Water Infrastructure Allocation

An unsupervised K-Means clustering algorithm has been utilised to translate pixel-level water stress prediction into a spatial planning output. This was performed on the 2035 water stress raster stack combining rainfall, groundwater depth, water stress intensity, population density, built-up density and soil infiltration index, in order to delineate zones of internally homogeneous physical and socio-hydrological character. Each of these zones were then matched to the most physically and operationally appropriate decentralized infrastructure type.

6.3.1 Optimal Number of Clusters

The Elbow Method-based WCSS curve exhibited a clear inflection at $K = 3$, beyond which the marginal reduction in within-cluster variance flattened substantially, indicating diminishing returns from additional clusters. Similarly, Silhouette Score also provided the highest value at $K = 3$ which suggests well-separated and internally coherent groups. Although Davis-Bouldin Index reached minimum at $K = 7$, it still had an acceptable low value for $K = 3$, which was chosen due to the following reasons:

- DBI-optimal $K = 7$ solution is not correspondingly supported by either the WCSS or Silhouette criteria, indicating that the $K = 7$ minimum is likely driven by the DBI's known sensitivity to small, tightly-bounded clusters.
- This study aims to classify decentralised infrastructural interventions into exactly three decentralized intervention categories: Recharge Wells, Community Water Storage Tanks, and Rainwater Harvesting Structures. Therefore, a three-cluster solution maps directly and interpretably onto the three infrastructure classes.

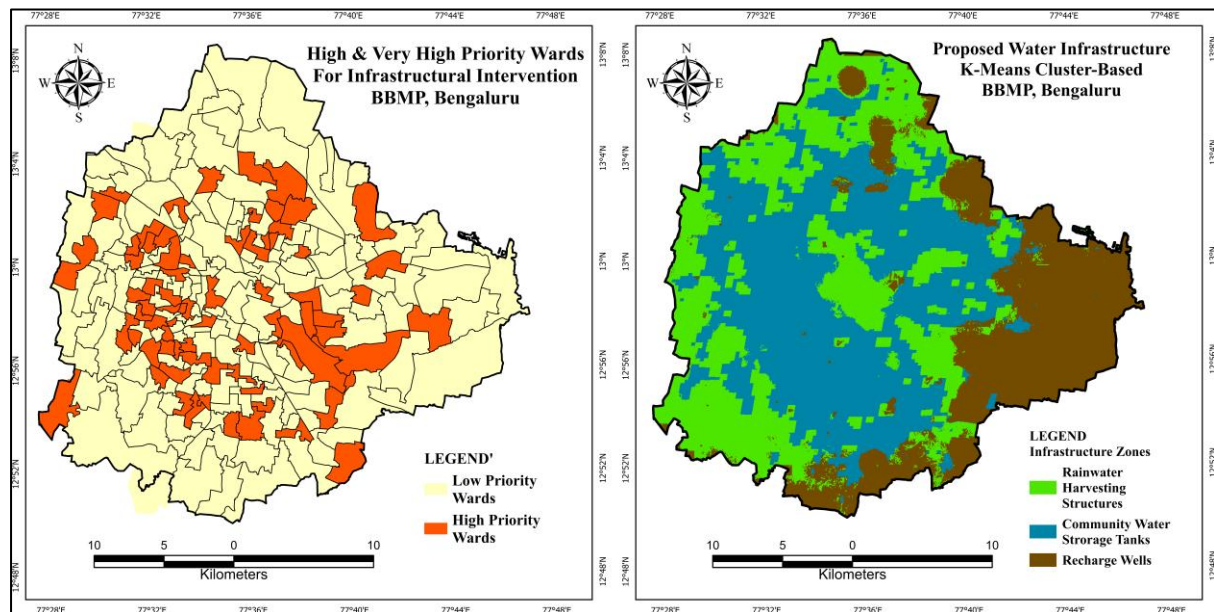
6.3.2 Cluster Characteristics & Infrastructures Allocation

Cluster 1 is being attributed to Recharge Wells as a decentralised infrastructure alternative. This cluster incorporates the following decisive characteristics: deep groundwater levels, high borewell density and high-to-moderate water stress intensity that critically coincides with locally higher soil infiltration potential relative to the built-up core. This cluster can be seen highly prominent in eastern and north-eastern wards of Bengaluru's BBMP. This combination of deep groundwater tables alongside high water infiltration capacity makes this cluster most suited for Recharge Well type decentralised intervention to combat water scarcity.

Cluster 2 encompasses the largest area in the decentralised infrastructure allocation map and it has been observed that this cluster spatially coincides with regions of high built up, dense

population and some of the highest water stress levels. Due to the presence of large-scale impervious surfaces coupled with low soil infiltration potential, this cluster is accordingly allocated Community Water Storage Tanks, which provide immediate, above-ground buffering and equitable distribution capacity to a dense resident population without depending on local infiltration capacity. This cluster therefore, can be seen in most central and western wards of Bengaluru.

Cluster 3 is largely situated in most northern, southern and south-western peripheral wards which are characterised by higher rainfall, lower population density and moderate-to-high soil infiltration potential, reflecting its more recently urbanising, less densely built character. Since it has been observed that these areas have comparatively highest amounts of predicted rainfall as well as high SPI and furthermore, these regions have comparatively lesser urban water stress, this cluster is therefore attributed to Rainwater Harvesting Structures at the household and community scale, which capture and store rooftop and surface runoff for direct use or shallow recharge, providing a lower-cost, decentralized supplementary supply appropriate to the still-developing infrastructure context of these wards ahead of the further population densification projected for 2035.



**Fig. 25: (a) High & Very High Water Stress Priority Wards of Bengaluru
(b) Optimal Decentralised Water Infrastructure Allocation Location of Bengaluru**

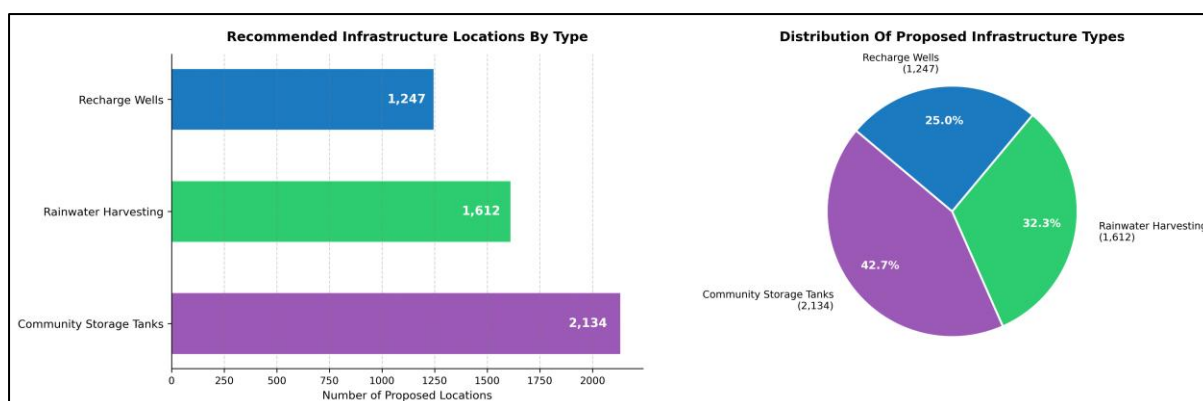


Fig. 26: Bar & Pie Chart showing Optimal Decentralised Infrastructure Distribution

6.3.3 Bengaluru's Ward Priority Distribution for Infrastructure Allocation

The Decentralised Infrastructure Zonation map is cross-referenced with Bengaluru's BBMP Ward map as well as Urban Water Stress map in order to delineate the "High" and "Very High" water-stressed wards of Bengaluru (Figure 25). 19 wards (8.5%) are identified as severely stressed with a further 49 wards (21.9%) as highly stressed. This accounts for almost one-third of Bengaluru's wards as high to severely high water stressed which is projected to further intense by 2035. Most of these high and very high stress wards are also present in "High Water Stress" zones of this study's machine-learning based Urban Water Stress map, which further validates the accuracy of the machine learning models. The remaining 110 "Moderate" priority wards (49.1%) and 46 "Low" priority wards (20.5%) are concentrated in the southern and far northern periphery, corresponding to the comparatively better-resourced or less-stressed zones identified throughout the preceding sections.

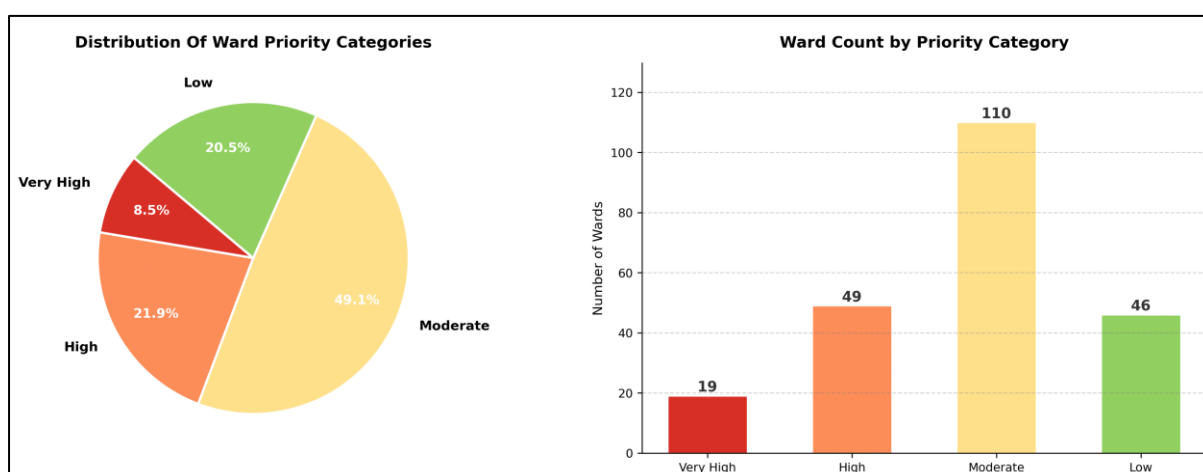


Fig. 27: Charts showing Ward Priority for Decentralised Infrastructure Allocation

CHAPTER 7:

Conclusion

This dissertation has sought to address the intensifying and multi-dimensional water crisis of Bengaluru through means of a reproducible GeoAI-enabled predictive framework to map and quantify the spatial distribution of urban water stress across the city and translate the findings into an actionable and sustainable water resource planning through proposal of optimal locations of future decentralised water infrastructural interventions, for long-term urban water resilience.

The study, therefore, has successfully adopted a multi-fold methodology including historical Land Use Land Cover (LULC) datasets, classified using supervised Random Forest for 2000, 2005, 2015, 2020 and 2025 as well as several statistical and stochastic models to predict future spatial distribution of several water stress drivers, that includes Facebook Prophet, ARIMA, CA-Markov as well as OLS Linear Regression. All models are validated through accuracy generation as well as R^2 and RMSE scores, which makes them valid and acceptable for processing. Unavailability of government dataset led to AHP-driven training samples generation as well as developing Random Forest and XGBoost-based water stress prediction models for 2025 and 2035, all of which consistently achieved high Overall Accuracy (%) above 85% and Near Perfect Kappa Agreement. The study confirms that high urban expansion, built-up acceleration coupled with groundwater depletion, borewell construction and shrinkage of urban lakes as the biggest driver of Bengaluru's water stress.

K-means based water infrastructure allocation and mapping has provided proposals regarding which areas of Bengaluru are suitable to Recharge Wells, Community Water Storage Tanks as well as Rainwater Harvesting Structures. The study further identified and quantified high and very high water-stress wards of Bengaluru that require immediate infrastructural interventions. Every finding achieved by machine learning based water stress prediction distribution as well as clustering algorithm has been externally corroborated against BWSSB groundwater level surveys, WELL Labs-based Water Security Index as well as an independent Water Policy study. This makes this study externally validated, cross-referenced and acceptable for further references.

The primary contribution of this research lies in its integration of forecasting, ensemble machine learning classification and unsupervised spatial clustering within a single GeoAI-driven decision-support framework, which is a combination that remains largely unexplored in existing Bengaluru-focused literature.

In conclusion, this study has demonstrated that Bengaluru's water stress is not a mere static or purely groundwater-driven phenomenon but a compounding, spatially expanding crisis shaped by rapid urbanisation, thermal intensification and infrastructural inequity.

7.1 Limitations of Study

- Absence of an officially validated ground-truth Urban Water Stress dataset for Bengaluru led to generation of training labels through AHP-based MCDA, which is a proxy for real-world observed stress rather than field-verified ground truth.
- The predictive framework extensively relied on secondary and remote sensing datasets of varying native spatial resolution. Therefore, resampling had to be performed to keep resolution consistent at 30 m which may have produced interpolation artefacts and mask certain variability in data.
- Borewell Density, Distance to Pumping Stations and existing Water Infrastructure were held constant between 2025 and 2035 due to the impracticality of forecasting future infrastructure development, which may be over or under-estimated.

7.2 Future Scopes of Work

- Integration of real-time IoT-enabled sensors on borewells, pumping stations and storage infrastructure may generate a continuous live updating Water Stress Index.
- Deep Learning methods like convolutional neural networks can be explored as they can capture more complex non-linear spatial dependencies among variables.
- Future studies may incorporate water quality parameters like groundwater salinity as well as arsenic or lead contamination to produce a more holistic Water Security Index.
- The study's Infrastructure Siting and Ward Priority methodology can be extended and validated for other rapidly urbanising metropolitan cities of India and the Global South which are also facing groundwater-induced urban water stress crisis.

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